

GENERAL SURGERY

Surgical Ethics

INTRODUCTION

Ethics - Definition

- The word ethics is derived from the Greek word “ethos”, which means “character”.
- Formally speaking, ethics is a branch of philosophy that defines things that are beneficial to individuals and society, and establishes the nature of obligations, or duties that people owe themselves and one another

Ethics - History

- The Greek healers in the 4th century BC drafted the Hippocratic Oath and promised to “prescribe regimens for the good of my patients according to my ability and my judgment and *never do harm* to anyone”.

Surgical Ethics

- Ethical study investigates what should be our character and conduct (behavior).
- Morality is subject to re-examination and improvement.
- The concepts of justice and fairness require critical evaluation and improvement.
- Ethical argument should remain relevant.

BASIC PRINCIPLES OF MEDICAL ETHICS

Autonomy

- When making decisions about healthcare procedures patients must have freedom of thought, intention, and action.
- Patients must be informed of the consequences of surgery that may adversely affect them.

Beneficence

- It means act of charity, mercy, and kindness
- The surgeon should act in the best interest of the patient.

- Surgeons depend on different technology, ranging from diathermy to the lighting in the operation room.
- Therefore, a conscientious (wishing to do one's work or duty well and thoroughly) surgeon should ensure that the equipment functions properly and reliably.
- Equipment failure can compromise patient care and increase the likelihood of surgical complications.

Non-maleficence

- It means non-harming or inflicting the least harm possible
- Most importantly, you should make sure not to do any harm.
- Ensure that the procedure doesn't harm the patient or any other individuals in society

Justice

- The allocation of scarce health resources and deciding who gets which resources.
- The four main areas that the surgeon must consider when assessing justice:
 1. Fair distribution of scarce resources
 2. Competing needs
 3. Rights and obligations
 4. Potential conflicts with established legislations

ISSUES – SURGICAL ETHICS

Autonomy

- You should respect the autonomy of patients and their ability to choose treatment options.
- Recognize the patient's right to self-determination.
- Patients have the right to choose their surgical treatments and surgical care.
- Respecting the autonomy of the patient is the basis for the ability of the patient to make informed consent.

Informed Consent and Difficulties

- Individuals have the right to obtain all available medical information and the ability to make autonomous decisions regarding their health care.
- Information that should be provided to the patient:
 1. Explanation of the patient's illness.
 2. Explanation of untreated natural history.
 3. Suggestion of the most suitable surgery.
 4. Discussion of the risks and the benefits.
 5. Alternative treatments and the expected outcome (Prognosis)

Consent Principles

- *Venue*: the venue should be a quiet and calm place.
- *Consent form*: the consent form should be in the patient's language.
- *Time*: you should take your own decision regarding the time.
- *Principal person*: you- the surgeon- are the principal person.
- *Entry*: case record.
- *The information*, which you provide to the patient, should be accurate and reasonably complete.
 1. You should avoid using technical language, to ensure that the patient understand what you say.
 2. Translators should be provided.
 3. You should clarify any doubts that the patient has.

Practical Difficulties

1. Refusal or waiver by patient.
2. Temporarily unconscious patients.
3. Children that are under the age of 18 are considered minors and are legally incompetent.
4. Incompetence of other kinds.

End-of-Life Issues

- In unusual (near to death) circumstances, where there is no evidence that a specific treatment that the patient desires is beneficial from any perspective, then the physician need not provide such treatment.
- If there is no treatment options, that is, the patient's brain is dead but the patient's family still insists on treatment, given that the physician is powerless (there is nothing that the physician can do), then the treatment must be stopped.
- Noted in case sheet along with the senior clinician's agreement.

Confidentiality

- The principle of confidentiality is that the patient's personal information, that is revealed to the surgeon by the patient is private, and has limits and restrictions on how and when it can be disclosed to a third party.
- The patient (and the treating doctor i.e. the surgeon), both have dignity.
- Breaking confidentiality (situations where breaking the confidentiality of the patient is acceptable):
 1. If the patient poses a threat to himself or to others in society.
 2. Sharing the information of the patient with other team members, to improve the treatment options.
 3. Public interest.
 4. Court order

Surgical Research

- Surgeons have the auxiliary (other) task to improve operative techniques to assure their patients that the care provided is best.
- The management of such regulation is conducted through research ethics committees, and surgeons are not allowed to participate in surgical research without the approval of such bodies.

Good Standards

- In order to successfully maintain life and health to an acceptable standard, surgeons must only provide specialized treatments, in which they have been properly trained.
- Thus, the surgeon must continue receiving education, throughout his entire career, in case of new surgical procedures.
- Not doing so is placing the interests of the surgeon higher than that of the patient, which is not acceptable, neither morally nor professionally.

Shock in Surgery

Shock is a state of acute cardiovascular or circulatory failure. It leads to decreased delivery of oxygenated blood to the body's organs and tissues or impaired oxygen utilization by peripheral tissues, resulting in end-organ dysfunction

PATHOPHYSIOLOGY OF SHOCK

Changes in preload, stroke volume, system vascular resistance, and cardiac output can result in impaired tissue and organ perfusion. The impaired delivery of oxygen to peripheral cells that occurs in shock results in a transition from aerobic to anaerobic cellular metabolism. Anaerobic metabolism generates lactate via metabolism of glucose to pyruvate, and lactate can be used as a surrogate marker for tissue hypoxemia and the severity of shock. Cells can engage in anaerobic metabolism for a limited time, but persistent cellular hypoxia results in cell death and tissue necrosis, leading to multi-organ system dysfunction and failure. The saturation of venous oxygen measured from central vessels (e.g. SVC), is another biochemical marker of peripheral O_2 uptake and can be used diagnostically to help in assessment of prognosis.

TYPES OF SHOCK

Hypovolemic Shock

It occurs due to inappropriately low intravascular volume leading to a decrease of preload, stroke volume, and cardiac output. Hypovolemic shock can be due to decreased intravascular fluid or decreased blood volume from hemorrhage for example.

Cardiogenic Shock

Myocardial infarction is the most common cause of cardiogenic shock, which results from failure of the left ventricle (LV) to generate adequate arterial flow to deliver oxygenated blood to peripheral tissues. Cardiogenic shock may be due to disruptions in stroke volume and/or heart rate. Failure of the LV may be due to right ventricle failure or valvular disease. Processes that can negatively affect stroke volume include aberrations in preload, afterload, and myocardial contractility.

Obstructive Shock

It results from either a critical decrease in preload or an increase in left ventricle outflow obstruction. Extra-cardiac processes that increase intra-thoracic pressure can result in obstructive shock by decreasing cardiac compliance and interrupting venous return by compressing the inferior or superior vena cava. Tension pneumothorax, herniation of abdominal contents into the thorax, and positive pressure ventilation are processes that result in decreased cardiac compliance and obstruction of the vena cava, decreased preload, and decreased cardiac output. Extra-cardiac processes that cause right ventricle outflow obstruction include severe pulmonary hypertension and massive pulmonary embolism. Increased right ventricle obstruction causes a decrease in right ventricle stroke volume, pulmonary arterial flow, left ventricle preload, and left ventricle cardiac output, as well as a decrease in delivery of oxygenated blood to peripheral tissue.

Anaphylactic Shock

Anaphylaxis can result in shock due to a mixed distributive and hypovolemic pathophysiology. Anaphylaxis results from activation of mast cells and basophils through immunoglobulin-E binding a specific allergen, resulting in the release of immuno-stimulatory and vasoactive proteins, with profound systemic vasodilation and diffuse vascular leak. Vasodilatation results in decreased systemic vascular resistance and mean arterial pressure (distributive pathophysiology). Vascular leak results in extravasation of intra-vascular fluid and decreased preload (hypovolemic pathophysiology). Volume resuscitation is appropriate in anaphylactic shock, but the mainstay of treatment is rapid administration of epinephrine, which should be given immediately if anaphylaxis is suspected. Histamine receptor antagonists (H1- and H2-blockers) and glucocorticoids are also recommended for patients with anaphylactic shock.

Septic Shock

It is defined as life-threatening organ dysfunction caused by a dys-regulated host response to infection. It is a subset of sepsis with circulatory and cellular/metabolic dysfunction associated with a higher risk of mortality. This can be caused by infections such as bacterial (Gram +ve and Gram –ve), fungal, viral and protozoal infections

CLINICAL FEATURES OF SHOCK

Manifestations of the cause of shock and features of shock depend on the degree of loss of volume and on duration of shock and can be classified into mild, moderate and severe as follows:

Mild Shock

- Collapse of subcutaneous veins of extremities esp. the feet, which become pale and cool
- Sweat on forehead, hand and feet
- Urine output normal.
- Pulse rate normal.
- Blood pressure normal.
- Patient feels thirsty and cold

Moderate Shock

- Mild shock features +
- Drowsy and confused
- Oliguria
- Pulse rate increased usually less than 100/min.
- Blood pressure normal initially then falls in later stage

Severe Shock

- Unconscious
- Gasping respiration
- Anuria
- Rapid pulse
- Profound hypotension

STAGES OF SHOCK

1. *Initial stage*: The cells become leaky and switch to anaerobic metabolism.
2. *Non-progressive (compensated) stage*: Attempt to correct the metabolic upset of shock.
3. *Progressive (decompensated) stage*: Eventually the compensation will begin to fail.
4. *Refractory stage*: Organs fail and the shock can no longer be reversed

MANAGEMENT OF SHOCK

Aim of Treatment

- Treat the cause
- Improve cardiac function
- Improve tissue perfusion

Principles of Resuscitation

A: Airway: patent upper airway

B: Breathing: adequate ventilation and oxygenation

C: Circulation: placement of adequate intravenous (IV) access

Emergency Measures and Monitoring

- Patients may need to be immediately intubated
- Central pulses should be palpated upon arrival
- The chest should be auscultated
- If the patient's blood pressure is unknown. Patients should be immediately placed on a cardiac monitor, and an automated blood
- Pressure cuff should be used to frequently monitor their mean arterial pressure.
- Intravenous (IV) access should be obtained, preferably with 2 large-bore IV lines.
- Occasionally, patients in shock will require an urgent central venous line placement for vascular access to deliver medications or for volume resuscitation. Intra-osseous line placement is a fast and viable central circulation access option in patients with difficult intravenous access.
- Oxygen saturation should be checked immediately and continuously monitored.
- An estimation of jugular venous pressure has been advocated to evaluate for right atrial pressures
- A 12-lead electrocardiogram (ECG) may help identify the etiology of shock; in addition to demonstrating evidence of myocardial infarction
- Urine output has to be assessed and catheterization is needed

Laboratory Investigations

- Complete blood count (CBC)
- Basic serum chemistries (including renal function), and other tests (liver function tests, lipase/amylase, cardiac biomarkers, etc)
- Arterial blood gas test. Venous blood gas test provides information of patient's acid/base status, but does *not* provide accurate information regarding oxygenation.
- Relevant cultures should be obtained early in the diagnostic work-up.
- Blood cultures are appropriate in most patients with suspected infection
- Urine, cerebral spinal fluid (CSF), pleural, ascitic, and/or other fluid compartment cultures should be sent, as clinically indicated

Imaging Studies

- X-ray chest
- Echocardiography
- Other studies related to the cause (ultrasound abdomen, computed tomography scan abdomen, etc)

Fluid Therapy in Shock

- *Crystalloid Solutions* (Normal saline - Ringers Lactate solution - Hartmann's solution): Crystalloids are typically indicated for the initial treatment of undifferentiated shock, although there is an emerging body of evidence that this may not be the best selection for trauma patients or patients with traumatic hemorrhagic shock

- *Colloid Solutions:*
Hydroxy-ethyl starch should not be used for volume resuscitation in patients in shock, especially septic shock
- *Blood transfusion*
RBCs contribute to oxygen carrying capacity (hemoglobin). Replacement with all other solutions will support volume and improve end-organ perfusion, but will not provide additional oxygen carrying capacity

Dynamic Fluid Response

Infusing 250-500 ml of fluids rapidly over 5-10 minutes

- Responders: Improvement
- Transient responders: revert back
- Non-responders

Vasopressors

Once a patient is determined to be euvolemic, but there is still ineffective oxygen delivery, vasoactive, various vasopressor medications may be used to support the mean arterial pressure by increasing systemic vascular resistance and/or cardiac output.

- *Nor-epinephrine* is a strong alpha agonist with some beta-1 activity, and it is a recommended. Initial choice for most categories of shock, particularly when the etiology of shock is unknown.
- *Inotropes (Dobutamine)* may be of value in cardiogenic shock / severe septic shock to increase the cardiac output, although their use has been linked in recent studies with higher incidence of arrhythmias

Antibiotic Therapy in Septic Shock

- Administration of IV anti-microbials should be initiated as soon as possible after recognition and within one hour for both sepsis and septic shock.
- Empiric broad-spectrum therapy with one or more anti-microbials is recommended for patients presenting with sepsis or septic shock to cover all likely pathogens.
- Empiric anti-microbial therapy should be narrowed once pathogen identification and sensitivities are established and/or adequate clinical improvement is noted.
- Systemic anti-microbial prophylaxis in patients with severe inflammatory states of non-infectious origin is *not* recommended.
- If combination therapy is initially used for septic shock, de-escalation is recommended with discontinuation of combination therapy within the first few days in response to clinical improvement and/or evidence of infection resolution

End-points of Resuscitation in Shock

Classic / Traditional

- Restoration of blood pressure
- Normalization of heart rate and urine output
- Appropriate mental status

Improved / Global

- All of the above plus
- Normalization of serum lactate levels
- Resolution of base deficit

Goal Directed Approach

- Urine output > 0.5 mL/kg/h
- Central venous pressure (CVP): 5 -10 cm H₂O
- Mean arterial pressure (MAP): 65-90 mmHg
- Central venous oxygen concentration > 70%

Hemostasis and Blood Transfusion

HEMOSTASIS

- Hemostasis is the act of restricting or stopping bleeding from a damaged vessel or injured organ during surgery or other invasive procedures.
- Bleeding may be of arterial or venous origin. Arterial bleeding is characterized by being pulsatile, whereas venous bleeding usually oozes without any pulsation.
- During surgery, balance should be maintained between bleeding and coagulation such that blood should continue to flow and supply tissues at the operative site while at the same time excessive blood loss is prevented/controlled

Factors that may Contribute to Bleeding in Surgery

Procedural Factors	Patient Factors
- Surgical incisions - Exposed bone - Large surfaces of exposed capillaries - Vascular adhesions stripped during surgery - Friable tissues not amenable for suturing	- Variant anatomical considerations - Medications (e.g. anticoagulants) - Coagulopathies (hepatic decompensation). - Platelet dysfunction or deficiency - Fibrinolytic activity - Poor nutritional status

Consequences of Uncontrolled Bleeding during Surgery

On the Patient	On the Surgeon/Procedure
- Anemia and hypovolemic shock. - Reduction in core temperature. - Thrombocytopenia. - The need for ICU admission - Longer postoperative hospital-stay - Hazards of blood transfusion and massive replacement.	- Prolongation of the procedure time - Obstruction of the surgical visual field.

Different Methods to Achieve Hemostasis

Methods used to achieve hemostasis are listed in the Table below¹. They may be mechanical, thermal, or chemical.

Table 1: Methods used to achieve hemostasis

Method	Examples
Mechanical Methods	1. Direct pressure 2. Fabric pads/gauze sponges/sponges 3. Sutures/staples/ligating clips
Thermal/Energy-Based Methods	1. Electro-surgery - Monopolar diathermy - Bipolar diathermy - Bipolar vessel sealing device - Argon enhanced coagulation 2. Ultrasonic device 3. Laser
Chemical Methods <i>Pharmacological Agents</i>	1. Epinephrine 2. Vitamin K 3. Protamine 4. Desmopressin 5. Lysine analogues (e.g. aminocaproic acid, transemic acid)
<i>Topical Hemostatic Agents</i>	1. Passive (ie, mechanical) agents - Collagen-based products - Cellulose - Gelatin - Polysaccharide spheres 2. Active agents: - Thrombin products 3. Flowables 4. Sealants - Fibrin sealants - Polyethylene glycol (PEG) polymers - Albumin and glutaraldehyde

Mechanical Methods

Direct Pressure

- The use of direct pressure or compression with one or more fingers at a bleeding site is typically a surgeon's first choice to attempt to control bleeding as is the simplest and fastest method (Figure 1). Arterial bleeding is more easily controlled with direct pressure than venous bleeding, which may not always be controlled with direct pressure, and in some cases, direct pressure may even increase the vascular injury and bleeding owing to its thin wall.



Figure 1. Hemostasis by direct pressure

- In general, maintaining pressure for 15 to 20 seconds will cause small clots to form at the end of blood vessels. If a major artery or vein has been injured, direct pressure should only be used as an initial temporary method to halt bleeding until the proximal and distal ends of the vessel become controlled or ligated using any of the methods mentioned below.

Fabric Pads/Gauzes/Sponge

- Sponge sticks are often used to apply pressure in deep body cavity recesses; care should be taken when removing the sponge stick to avoid dislodging fresh clots.
- Packing an area of venous bleeding can help to reduce blood loss when direct pressure control is not an option or when there is generalized bleeding from systemic coagulopathy that has occurred because of infection, trauma, massive blood loss, or platelet dysfunction. Counting the number of sponges used in packing of a cavity should be performed by the surgical team member placing them to ensure all items used will be retrieved before wound closure to avoid missed or retained surgical pack

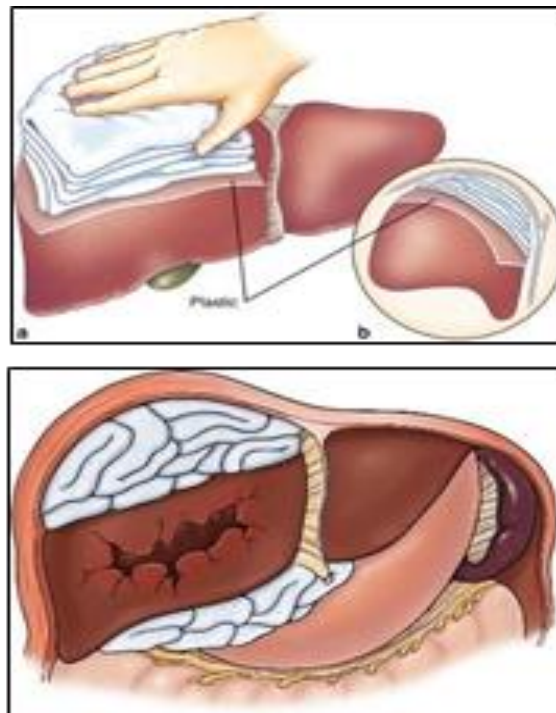


Figure 2. Hemostasis by packing the bleeding area

Sutures/Staples/Ligating Clips

- Sutures and ties are used during operative procedures as ligatures to tie off blood vessels and control bleeding (Figure 3). Considerations when using suture include the type of tissue it will be used on, its tensile strength, and whether it is absorbable or not.
- Staples are applied by using sterile, disposable stapling devices that place staggered rows of titanium staples and concurrently divide the tissue located between the rows of staples to achieve hemostasis (Figure 4).
- Ligating clips or hemostatic clips are used to control bleeding blood vessels in an easy and quick way to achieve hemostasis efficiently and reduce the risk of foreign body reaction that may occur with suture material. Ligating clips are available in various sizes and must be used with the corresponding size clip applicator (Figure 4).

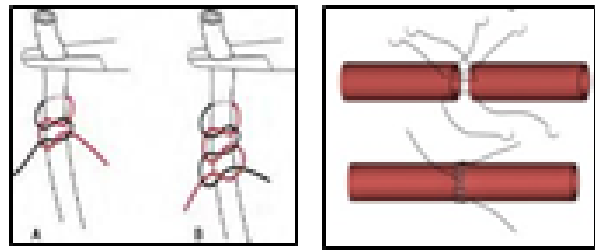


Figure 3: Hemostasis by sutures intra-operatively

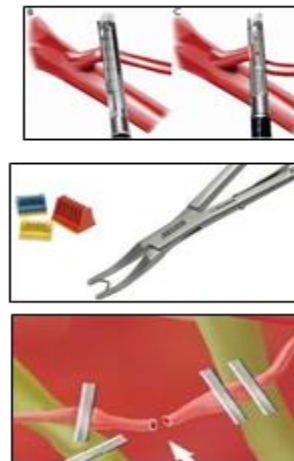


Figure 4: Hemostasis by applying staples or clips

Thermal/Energy-Based Methods

Electro-surgery

Electrocautery was developed in the 1930s and it entails the use of high-frequency alternating current for cutting, coagulating, and vaporizing tissues in both open and laparoscopic surgical procedures. Electrosurgical energy is delivered in either monopolar or bipolar mode. A complete electrical circuit is required for current to flow when using either of these modes. The potential risks of electro-surgery use include patient injuries, operating surgeon(s) injuries, fires, and electromagnetic interference with other medical equipment or in-vitro electronic devices. Accordingly, to use such devices, safety is heightened by adhering to good clinical practices, and awareness of adverse events that may occur with patient burns.

1) Monopolar electro-surgery

It is the most frequently used electrosurgical method of hemostasis. In a monopolar circuit, electrical current flows from the generator through an active electrode to the patient where controlled heat is produced, which results in either cutting or coagulation of the tissue. The completion of the circuit requires the current to flow from the patient to the dispersive electrode placed on the patient's body before it returns to the generator.

2) *Bipolar electro-surgery*

It is another mode where the electrical circuit is completed between the two blades of a bipolar forceps without the need to flow through the patient. Accordingly, a dispersive electrode is not needed. Heat is generated in the tissue as the current flows from one bipolar tip through the tissue and back to the other tip (Figure 5).

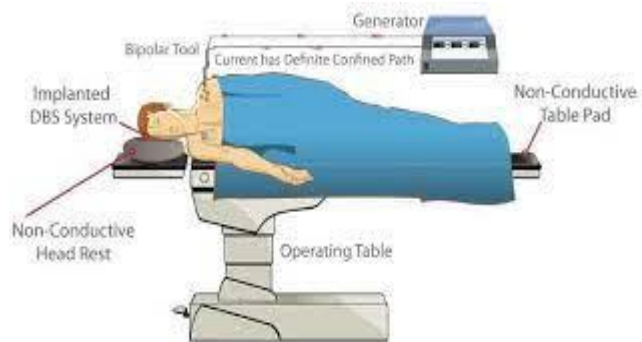


Figure 5: Hemostasis by bipolar electrosurgery

3) *Bipolar vessel sealing technology*

It is an advanced electrosurgical modality in which the intimal layers of a vessel are fused together creating a permanent seal (Figure 6). Such devices apply heat over time with high compression, which makes them capable of simultaneously sealing and transecting vessels up to 7 mm in diameter, large tissue pedicles, or vascular bundles.

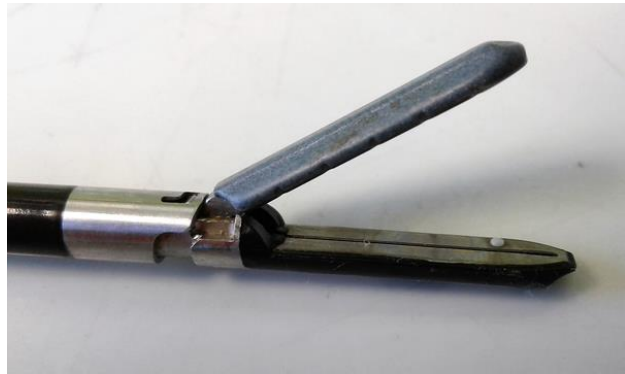


Figure 6: Hemostasis by bipolar vessel-sealing

4) *Argon-enhanced coagulation*

Argon-enhanced coagulation technology uses a stream of inert, non-combustible argon gas to conduct the electrosurgical current (Figure 7). Argon gas is heavier than air and displaces nitrogen and oxygen. The electrosurgical current ionizes the argon gas, which makes it more conductive to heat than air and helps create a bridge between the electrode and the tissue.



Figure 7: Hemostasis by Argon plasma coagulation

Lasers

Laser the acronym for *light amplification by stimulated emission of radiation*. They are common heat-generating devices used by surgeons to provide hemostasis.

Ultrasonic Devices

Ultrasonic devices represent another example of vessel-sealing devices that convert electrical energy into mechanical energy that oscillates longitudinally at the point of contact of both device tips enabling simultaneously coagulation and cutting of the vessel ends (Figure 8). These devices can seal vessels up to 5 mm in diameter and offer an alternative to electrosurgical devices mentioned above.



Figure 8. Hemostasis by ultrasonic device

Chemical Methods/Topical Hemostatic Products

Pharmacological Agents

- *Epinephrine* causes direct vasoconstriction of blood vessels making it useful during surgery, because it can be applied topically to reduce bleeding.
- *Vitamin K* also plays a role in the coagulation process and may be administered preoperatively as a pro-coagulant to reverse the effects of warfarin. It should be given orally or IV rather than subcutaneous or intramuscular injection route.
- *Protamine* is the only current agent to reverse heparin anticoagulation. (But not low-molecular-weight heparin).
- *Desmopressin* is a synthetic analogue of vasopressin. It stimulates the release of von Willebrand factor (vWF) from endothelial cells to enhance primary hemostasis. While it assists in reducing perioperative bleeding, the effect of desmopressin is too small to influence other, more clinically relevant outcomes, such as the need for blood transfusion and repeat procedures.
- **Synthetic lysine analogues** are anti-fibrinolytic agents that competitively inhibit activation of plasminogen, thereby reducing its conversion to plasmin, and thus reducing the degradation of fibrin clots.

Topical Hemostatic Products

A wide range of topical hemostatic products are available for use in the operating room. Such topical agents act to achieve hemostasis either passively or actively.

Passive hemostatic agents

- These hemostatic agents achieve control of bleeding via acceleration of the coagulation cascade by forming a physical, lattice-like matrix that adheres to the bleeding site which thereby activates the extrinsic clotting pathway and provide a platform around which platelets can adhere and aggregate to form a fibrin clot. Accordingly, these agents can only be used among patients with intact coagulation profile. They are generally used as first-line hemostatic agents since they are usually readily available, require no special storage or preparation, and relatively inexpensive.

- Available passive hemostatic agents include:
 1. Collagen-based products as Microfibrillar® and Ultra-foam® (non-absorbable), or Helistat® (absorbable).
 2. Oxidized regenerated cellulose such as Surgicel® and Surgicel Nu-Knit®.
 3. Porcine gelatins such as Gelfoam®.

Active hemostatic agents

- Active hemostatic agents, in contrast, have biological activity and directly participate at the end of the coagulation cascade to stimulate conversion of fibrinogen at the bleeding site to fibrin to create a fibrin. Active hemostatic agents include topical thrombins (e.g. bovine, pooled human plasma, or recombinant). Since thrombin acts at the end of the clotting cascade, its action is less affected by coagulopathies from clotting factor deficiencies or platelet malfunction, and thus useful for patients receiving antiplatelet and/or anticoagulation medications.
- However, topical thrombin requires the presence of fibrinogen in the patient's blood to be able to act which makes it ineffective in patients having afibrinogenemia. Hemostasis typically occurs within 10 minutes in most patients and the control of local bleeding via active hemostatic agents is more effective than passive agents, which makes them more expensive.

Flowable hemostatic agents

- Flowable hemostatic agents combine passive and active hemostatic agents into a single application product (e.g., FloSeal®).

Sealants

- Sealants work by forming a barrier that is impervious to the flow of most liquids.
- There are four types of materials approved for use as sealants for surgical hemostasis:
 1. Fibrin sealants
 2. Polyethylene glycol (PEG) polymers
 3. Albumin with glutaraldehyde
 4. The new cyanoacrylate sealant.

BLOOD TRANSFUSION

- Blood transfusion is an essential therapeutic intervention that has been widely used in medical practice to manage anemia or hemorrhage since the early 20th century.
- Blood transfusion aims at providing recipients with blood component(s) to improve their physiological status.

Indications of Blood Transfusion

Acute Anemia

- Surgical hemorrhage (Hgb < 8 g/dl or presence of symptoms)
- Traumatic hemorrhage complicated by shock with inadequate tissue perfusion

- Acute non-surgical/non-traumatic hemorrhage as bleeding peptic ulcer or bleeding diverticular disease (Hgb <7 g/dl or presence of symptoms).
- Critical illness with Hgb <7 g/dl or presence of symptoms.
- Septic shock with Hgb <7 g/dl.

Chronic anemia

- Chronic blood loss (hepatic disorders, bleeding disorders).

Blood Groups

There are four types of blood groups, O, A, B, and AB. Blood typing test is used to find out the blood group among patients. The type of blood group is determined by the presence of certain type of *antigens* on the red blood cells (RBCs).

- Group A has "A" antigen
- Group B has B antigen
- Group AB has both A and B antigens
- Group O has core H antigen (universal donor).

Our serum naturally has antibodies against the red cell antigen that is *not* present on our own RBCs. Accordingly:

- Group A has anti-B antibodies
- Group B has anti-A antibodies
- Group AB has no anti-A or –B antibodies (universal recipient)
- Group O has both anti-A and anti-B antibodies

This creates the basis for blood typing and cross-matching of donor's and recipient's blood prior to transfusion (Table).

Blood Groups (AB, A, B, O)

Blood Group	Antigens on RBCs	Antibodies in serum to	Can donate RBCs to:	Can receive RBCs from:
AB	A and B	None	AB	A, B, AB, O
A	A	Anti B	A, AB	A, O
B	B	Anti A	B, AB	B, O
O	None	Anti A and anti B	O, A, B, AB	O

Available Types of Blood Products

1. Whole Blood

- It is the unseparated blood collected into a container containing an anticoagulant-preservative solution.
- Indicated for RBC replacement associated with hypovolemia as in hemorrhagic shock.

2. Red Cell Concentrate / Packed RBCs

- It comprises RBCs after centrifugation of whole blood and removal of most of plasma.
- Indicated for replacement of red cells in anemic patients without hypovolemia.

3. Platelet Concentrate

- Indicated for prevention or treatment of bleeding due to thrombocytopenia or platelet dysfunction.

4. Fresh Frozen Plasma (FFP)

- It is plasma that has been separated from a unit of whole blood.
- It contains water, electrolytes, clotting factors and other proteins (mostly albumin).
- Indicated for enhancing coagulation by restoring plasma clotting factors as in chronic severe liver diseases, DIC, and anti-thrombin III deficiency.

5. Cryoprecipitate

- It is the cold insoluble precipitate component of blood having factor VIII, vWF, fibrinogen and factor XIII as its major constituents.
- Indicated for treating factor VIII Deficiency (Hemophilia A), Von Willebrand's Disease, Hypo-fibrinogenemia (as in disseminated intra-vascular coagulation [DIC], dysfibrinogenemia, or afibrinogenemia).

6. Plasma Derivatives

- Sterile, concentrated specific proteins obtained from large pools of donor plasma.
- They include albumin and immunoglobulins.
- They are indicated in treating patients with specific protein deficiencies or to provide them with passive immunity

Complications of Blood Transfusion

- Acute hemolytic reaction due to mis-matched transfusion.
- Urticarial rash.
- Anaphylaxis.
- Non-hemolytic febrile reaction due to cytokines from the transfused blood unit released into the recipient's circulation.
- Transfusion-related acute lung injury.
- Sepsis due to infected blood product.
- Circulatory overload.
- Air embolism.
- Iron overload.

Wounds and Wound Healing

Definition

- A wound is an injury, especially one in which the skin or another external surface is torn, pierced, cut, or otherwise broken.

OPEN WOUNDS

Etiology (Types)

1. Incisions or Incised Wounds

These are caused by a clean, sharp-edged object such as a scalpel (surgical wound), a razor or glass-splinter.

2. Lacerations

Irregular tear-like wounds caused by some blunt trauma. Lacerations & incisions may appear linear (regular) or stellate (irregular). The term *laceration* is commonly misused in reference to incisions.

3. Abrasions (Grazes)

Superficial wounds in which the top most layer of the skin (epidermis) is scraped off. Abrasions are often caused by a sliding fall onto a rough surface.

4. Puncture Wounds

These are caused by an object puncturing the skin (e.g. nail or needle).

5. Penetration wounds

These are caused by an object such as a knife entering and coming out from the skin, or reaching a body cavity.

6. Gunshot wounds

Gunshot wounds are caused by a bullet or similar projectile driving into or through the body. There may be two wounds, one at the site of entry & one at the site of exit, generally referred to as a "*through-and-through*."

Classification of Surgical Wounds

As shown in Table 1, wounds may be considered as “clean”, “clean-contaminated”, “contaminated”, or “dirty”.

Table 1: Classification of Surgical Wounds

Type of Wound	Characteristic Features
1. <i>Clean</i>	- No hollow viscus entered - Primary wound closure - No inflammation - No breaks in aseptic technique - Elective procedure
2. <i>Clean-contaminated</i>	- Hollow viscus entered, but controlled - Primary wound closure - No inflammation - Minor breaks in aseptic technique - Mechanical drain used - Bowel preparation pre-operatively
3. <i>Contaminated</i>	- Uncontrolled spillage from viscus - Inflammation apparent - Major breaks in aseptic technique
4. <i>Dirty</i>	- Untreated and uncontrolled spillage from viscus - Pus in the operative wound - Open suppurative wound, severe inflammation

CLOSED WOUNDS

Etiology

The types of closed wounds are:

1. ***Contusions***

They are more commonly known as *bruises*. They are caused by a blunt force trauma that damages the tissues under the skin.

2. ***Hematomas***

They are caused by damage to a blood vessel that in turn causes blood to collect under the skin.

3. ***Crush injury***

They are caused by a great amount of force applied for a long time.

PHYSIOLOGY OF WOUND HEALING

A surgeon's role in wound management is to create an environment in which the healing process can proceed in an optimal fashion. As noted by *John Hunter*, "injury alone has in all cases a tendency to produce the disposition & the means of a cure."

History

- The earliest accounts of wound healing date back to about 2000 B.C.
- **Galen** emphasized the importance of maintaining a *moist* environment to ensure adequate healing.
- **Ambrose Paré** found that simply dressed gunshot wounds heal faster & are less painful than when treated with boiling oil, the previously accepted method.
- **Ignaz Philip Semmelweis** advocated need for *washing hands*
- **Joseph Lister** began soaking his instruments in phenol & spraying the operating rooms, reducing the mortality rates (MR) from 50% to 15%.

Processes of Wound Healing

Repair of tissue damage is separated into two processes, regeneration and healing.

1. *Regeneration* refers to growth of cells and tissues to replace lost structures.
2. *Healing* is the effort of tissues to restore normal function and structure after injury
 - To reform barriers to fluid loss and infection
 - To limit further entry of foreign organisms and material.
 - To re-establish normal blood and lymphatic flow patterns.
 - To restore the mechanical integrity of the injured system.

Primary Mediators of Wound Healing

- Wound healing process is orchestrated by the carefully regulated release of *growth factors* (GFs) and *cytokines*
- GFs bind to specific receptors on cells that deliver signals that have 2 main effects
 1. Stimulation of transcription
 2. Regulation of cell entry into the cell cycle

Growth Factors (GFs)

- These are polypeptides produced in normal and wounded tissues that stimulate cellular migration, proliferation and function (Table 2).
- Often named for the cells from which they were first derived. However, names are often misleading, because GFs have been shown to have multiple functions.
- Most GFs are very potent and cause significant effects in *nanomolar* concentrations (Table 2).

Table 2: Growth factors: cell origin and biological effects

Growth Factor	Wound Cell Origin	Cellular and Biologic Effects
PDGF	Platelets, macrophages, monocytes, smooth muscle cells, endothelial cells	Chemotaxis: fibroblasts, smooth muscle, monocytes, neutrophils Mutagenesis: fibroblasts, smooth muscle cells: Stimulation of angiogenesis, collagen synthesis
FGF	Fibroblasts, endothelial cells, smooth muscle cells, chondrocytes	Angiogenesis Mitogenesis: mesoderm and neuroectoderm fibroblasts, keratinocytes, chondrocytes, myoblasts
Keratinocyte growth factor	Keratinocytes, fibroblasts	Significant homology with FGF; stimulates keratinocytes
EGF	Platelets, macrophages, monocytes (also identified in salivary glands, duodenal glands, kidney, and lacrimal glands)	Stimulates proliferation, migration, and differentiation of all epithelial cell types
TGF	Keratinocytes, platelets, macrophages, T lymphocytes, neutrophils, monocytes	Homology with EGF; binds to EGF receptor, Mitogenic and chemotactic for epidermal and endothelial cells Stimulates angiogenesis TGF β_1 stimulates wound matrix, β_3 inhibits scar formation
Granulocyte-macrophage colony-stimulating factor	Macrophage/monocytes, endothelial cells, fibroblasts	Stimulates macrophage differentiation/proliferation
Insulin-like growth factors (IGF-I, IGF-II)	Platelets (IGF-I in high concentrations in liver; IGF-II in high concentrations in fetal growth); likely the effector of growth hormone action	Promote protein/extracellular matrix synthesis Increase membrane glucose transport
Vascular endothelial growth factor	Macrophages, fibroblasts, keratinocytes	Similar to PDGF Mitogen for endothelial cells (not fibroblasts)

Cytokines

- These are small proteins or glycoproteins secreted for the purpose of altering the function of target cells in an endocrine, paracrine, or autocrine fashion.
- They are *pleiotropic* i.e. have multiple effects from a single gene (Table 3)

Table 3: Cytokines: cell origin and biological effects

Cytokine	Wound Cell Origin	Cellular and Biological Effects
TNF-α	Macrophages	- PMN migration and cytotoxicity - Collagen synthesis - Provides metabolic substrate
IL-1	Macrophages Keratinocytes	- Fibroblast and keratinocyte chemotaxis - Collagen synthesis
IL-2	T-Lymphocytes	- Increase fibroblast infiltration and metabolism
IL-6	Macrophages PMNs Fibroblasts	- Fibroblast proliferation - Hepatic acute-phase protein synthesis
IL-8	Macrophages Fibroblasts	- Fibroblast, PMN and keratinocyte chemotaxis
IFN-γ	T-Lymphocytes Macrophages	- Macrophage and PMN activation - Retards collagen synthesis and cross-linking - Stimulates collagenase activity
Anti-Inflammatory Cytokines		
IL-4	T-Lymphocytes Basophils Mast cells	- Inhibition of TNF, IL-1 and IL-6 production; fibroblast proliferation, collagen synthesis
IL-10	T-Lymphocytes Macrocytes Keratinocytes	- Inhibition of TNF, IL-1 and IL-6 production. - Inhibits macrophage and PMN activation

TNF: tumor necrosis factor, IL: interleukin, IFN: interferon, PMN: Polymorphonuclear neutrophil

Phases of Wound Healing

Normal wound healing follows a predictable pattern that can be divided into overlapping phases defined by characteristic cellular populations and biochemical activities (Figure 1).

1. Hemostasis and inflammation
2. Proliferation
3. Maturation and remodeling

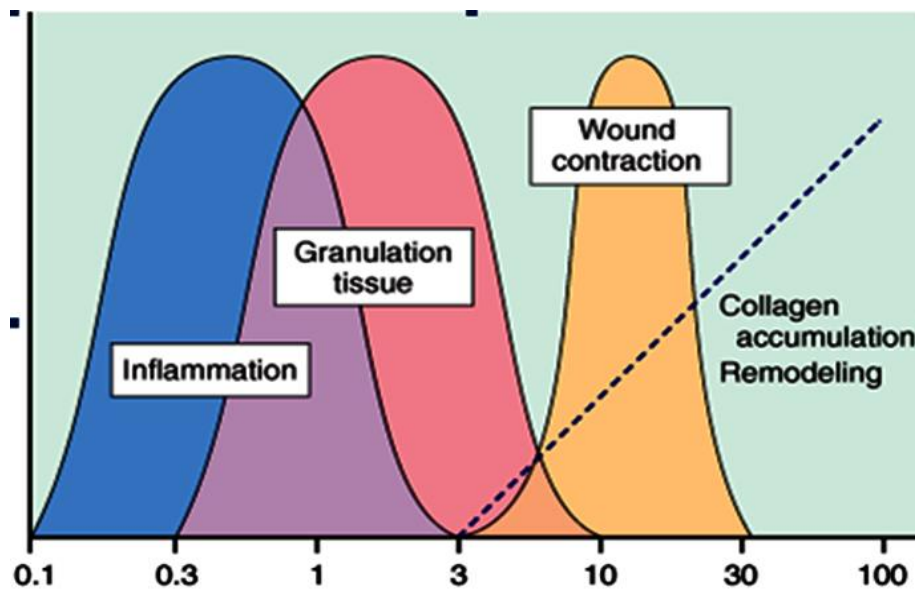


Figure 1. Three phases of wound healing

Inflammatory Phase

It represents the tissue's attempt to limit damage and is closely related with healing process. Healing is impossible without inflammation. The events can be divided into vascular events and cellular events

Vascular Events

- The vascular events include:
 1. Earliest manifestation is vasodilatation (VD), which follows a transient constriction of arterioles lasting a few seconds.
 2. Wounding disrupts tissue integrity and direct exposure of extra-cellular matrix to platelets.
 3. Initial contact between platelets and collagen requires *vonWillebrand factor* (vWF).
 4. The clotting cascade is initiated through both intrinsic and extrinsic pathways.
- Vasodilatation is followed by increased permeability of micro-vasculature, followed by stasis, which → accumulation of leucocytes along the vascular endothelium, which then migrate through the vascular wall into interstitial tissue.
- Permeability increases due to
 1. Formation of endothelial gaps in venules,
 2. Direct endothelial injury,
 3. Delayed prolonged leakage,
 4. Leucocyte mediated endothelial injury,
 5. Increased transcytosis and leakage from new vessels
- The combination of intense VD and increased vascular permeability → clinical findings of inflammation; *rubor* (redness), *tumor* (swelling), *calor* (heat) and *dolor* (pain).

Cellular Events

- Cellular infiltration after injury follows a characteristic, predetermined sequence as shown in Figure 2.
- *Inflammatory Granulocytes*
 - These are attracted by inflammatory mediators in the first 48 hours
 - They produce oxygen-derived free radicals
 - Non-specific

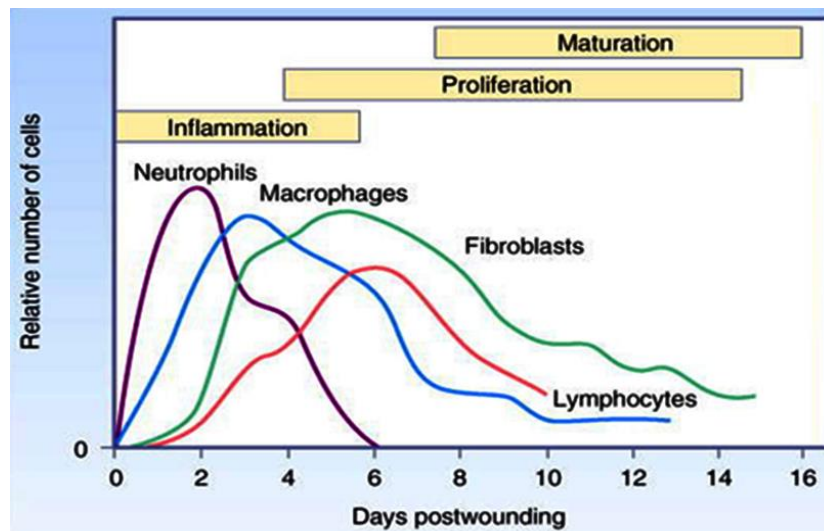


Figure 2.
Cellular events after wounding

- *Inflammatory Macrophages*
Monocytes
 - Attracted to the area by complement. Activated by: fibrin, foreign body (FB) material and exposure to hypoxic & acidotic environment
 - Reached maximum after 24 hours
 - Remain for weeks
- *Activated Macrophages*
 - *Essential* for progression onto the proliferative phase
 - Mediate (1) Angiogenesis (FGF, PDGF, TGF- α & β and TNF- α) and (2) Fibroplasia (IL's, EGF & TNF)
 - Synthesize NO
 - Secrete collagenases
- *T-Lymphocytes*
 - Population of inflammatory/immune cells that routinely invades the wound.
 - Less numerous than macrophages, numbers peak at about 1 week post injury
 - Bridge the transition from inflammatory to proliferative phase of healing
 - Depletion of most wound T-lymphocytes reduces wound strength and collagen content
 - Also exert a down regulating effect on fibroblast collagen synthesis by cell-associated interferon- γ , TNF- α & IL-1.

Proliferative Phase

- It is the second phase of wound healing and roughly spans days 4 through 12
- It is during this phase that tissue continuity is re-established
- Fibroblasts and endothelial cells are the last cell populations to infiltrate the healing wound & the strongest chemotactic factor for fibroblasts is PDGF.
- Recruited fibroblasts first need to proliferate & then become activated, to carry out their 1st function of matrix synthesis remodeling.
- The 4 major events are: *fibroplasia, angiogenesis, epithelialization and contraction*

1. *Fibroplasia*

- The proliferative phase begins with degradation of the initial fibrin-platelet provisional matrix.
- During fibroplasia, fibroblasts synthesize and deposit the replacement ECM at the wound site
- As fibroblasts proliferate, they become predominant cell types by 3-5 days in clean, non-infected wounds.
- The initial fibrin matrix is replaced by a provisional matrix of *fibronectin & hyaluron*, which facilitates fibroblast migration.
- Fibrous structural proteins such as *collagen and elastins*

Collagen

- It is the most common protein in the animal world providing the extracellular framework for all multicellular organisms
- Its deposition, maturation, and subsequent remodeling are essential to the functional integrity of the wound.
- At least 27 types of collagen encoded by 41 genes dispersed on at least 14 chromosomes,
- *Type I collagen* is the major component of extracellular matrix in skin.
- *Type III*, normally present in skin, becomes more prominent and important during the repair process
- *Type IV* is non-fibrillar, main component of basement membrane together with laminin.
- Each chain of collagen is composed of a glycine residue in every 3rd position. The 2nd position in the triplet is made up of *Proline* or *Lysine*

Elastin, Fibrillin and Elastic Fibers

- Provide the resilience to allow for recoil after transient stretch.
- Elastin is composed of hydrophobic & alanine- and lysine-rich α -helical segments that alternate along the polypeptide chain
- Elastic fibers consist of an elastin core covered with a sheath of microfibrils, which are composed of several distinct glycoproteins, such as fibrillin.
- Microfibrils appear before elastin in developing tissues and seem to form a scaffold on which the secreted elastin molecules are deposited.

2. **Angiogenesis**

- It is the process of new blood vessel (BV) formation
- *Macrophages* regulate angiogenesis during the inflammatory phase.
- Angiogenesis occurs by
 - a. Degradation of the basement membrane of post-capillary venules
 - b. Migration of cells through this gap is promoted by FGF, PDGF and TGF- β . PECAM-1, also found on endothelial cells, modulates their interaction with each other as they migrate into the wound
 - c. Tubule or lumen formation involving cell-cell and cell-matrix interactions. New capillaries differentiate into arterioles and venules, whereas others undergo involution and apoptosis, with ingestion by macrophages.
 - d. Deposition of the basement membrane resulting in capillary maturation.

3. **Epithelization**

- New epithelial cells for wound closure are provided by fixed basal cells in a zone near the edge of the wound
- The epidermal cell layer thickens and the marginal basal cells migrate over the wound defect. Once these keratinocytes begin migrating they do *not* divide until epidermal continuity is restored.
- Daughter cells flatten and migrate over the wound as a sheet, moving in a leap-frog and tumbling fashion (epiboly).
- Migration of keratinocytes over the wound is guided by cell adhesion glycoproteins, such as tenascin & fibronectin, which are their “railroad tracks”.
- After re-establishment of the epithelial layer, keratinocytes and fibroblast secrete laminin and type IV collagen to form basement membrane
- Keratinocytes become columnar and divide to restore the layering of the epidermis and reform a barrier.

4. **Contraction**

- It is the process in which the surrounding skin is pulled circumferentially toward an open wound.
- It does *not* occur with closed surgical incisions.
- It reduces the size of the wound dramatically without new tissue formation.
- Speeds wound closure compared to epithelization and scar formation alone
- The amount of contraction is related to both the size of the wound and the mobility of the skin.
- In humans contractions is greatest in the trunk and perineum, least on the extremities and intermediate on the head and neck.

Remodeling Phase

- The ECM is dynamic and is constantly undergoing remodeling
- Collagen cross-linking $\downarrow$ its degradation and improves wound tensile strength
- *Lysyl oxidase* is the major intermolecular cross linking enzyme.
- Degradation is by collagenases, gelatinases and matrix metallo-proteinases

- Scar formation is the final outcome of wound repair in children and adults
- The ultimate pattern of collagen in scar is one of densely packed fibers and *not* the reticular pattern found in unwounded dermis
- Scar has no epidermal appendages (hair follicles & sebaceous glands) and it has a collagen pattern that is distinctly different from the unwounded skin
- Remodeling occurs during months to years to form a mature scar.
- The early scar appearance is red due to its dense capillary network induced at the injury site
- Scars are usually hypo-pigmented after full maturation.
- However, it can become hyper-pigmented in darker pigmented patients and in those lighter pigmented patients who receive sun exposure.
- During remodeling, wounds gradually become stronger with time.
- Wound tensile strength increases rapidly from 1-8 weeks after wounding and correlates with collagen cross-linkage.
- It increases at a slower pace till 1 year.
- Tensile strength of the wound best reaches only 80% that of unwounded skin.

FACTORS AFFECTING WOUND HEALING

Factors affecting wound healing include age, infection, nutrition, hypoxia, anemia, hypo-perfusion, metabolic disorders, steroids, chemotherapeutic drugs and ionizing radiation

Age

- Aging produces intrinsic physiological changes that lead to delayed or impaired wound healing.
- With aging, collagen undergoes qualitative and quantitative changes.
- Dermal collagen content decreases with aging and aging collagen fibers show distorted architecture & organization.
- The increased incidence of cardiovascular disease, metabolic diseases (diabetes mellitus, malnutrition and vitamin deficiencies), cancer all contribute to the higher incidence of wound problems in the elderly

Infection

- It is the most common cause of healing delays
- If the bacterial count in the wound exceeds 10^5 organisms per gram of tissue, or if any β -hemolytic streptococci are present, the wound will *not* heal by any means.
- Bacteria prolong the inflammatory phase and interfere with epithelialization, contraction, and collagen deposition.
- Endotoxins stimulate phagocytosis and release of collagenase
- Bacteria may accelerate expression or increase concentrations of MMPs, GFs and cytokines in chronic-type wounds.

- Inactive precursors of MMPs are activated by bacterial proteinases of the Thermolysin family (*Pseudomonas*, *Vibrio* and *Serratia*).
- Bacterial phospholipase C can disrupt normal re-epithelialization by decreasing cell-cell contact and increasing cell migration, possibly by altering integrin expression and by upregulating MMP-9.

Nutrition

- Precise caloric requirements for optimal healing have *not* been determined.
- Malnourished patients have diminished *hydroxyproline* accumulation (an index of collagen deposition) into subcutaneously implanted polytetrafluoroethylene tubes when compared to normally nourished patients.
- Malnutrition correlates clinically with enhanced rates of wound complications and increased wound failure after diverse surgical procedures.

Nutrition - Arginine

- Arginine deficiency results in decreased wound-breaking strength and wound collagen
- The main effect of Arginine on wound healing is to enhance wound collagen deposition.
- Reduction in breaking strength during the first weeks of healing are directly related to new collagen synthesis
- Arginine supplementation may lead to improvement in wound strength as a consequence of enhanced collagen deposition

Nutrition - Vitamin A

- Deficiency impairs wound healing, whereas supplemental vitamin A benefits wound healing in non-deficient humans and animals.
- Vitamin A increases the inflammatory response in wound healing, probably by increasing the lability of lysosomal membranes.
- There is increased influx of macrophages, with increase in their activation and increased collagen synthesis.
- Directly increases collagen production and epidermal GF receptors when it is added *in-vitro* to cultured fibroblasts.
- Supplemental vitamin A can reverse the inhibitory effects of corticosteroids on wound healing.

Nutrition – Scurvy (Vitamin C deficiency)

- It causes a defect in wound healing, particularly via a failure in collagen synthesis & cross-linking.
- Vitamin C is required for the conversion of Proline and Lysine to Hydroxyproline and Hydroxylysine, respectively.
- Vitamin C deficiency is also associated with increased incidence of wound infection

Nutrition – Zinc

- In deficiency states, there is reduced fibroblast proliferation, reduced collagen synthesis, impaired overall wound strength and delayed epithelialization.

Hypoxia, Anemia and Hypoperfusion

- Low O₂ tension has a deleterious effect on all aspects of wound healing.
- Fibroplasia is significantly impaired by local hypoxia.
- Optimal collagen synthesis requires O₂ as a co-factor
- Factors affecting local O₂ delivery are either *systemic* (low volume or cardiac failure) or *local* (arterial insufficiency, local vasoconstriction, or excessive tension on tissues).
- The level of vasoconstriction of the subcutaneous capillary bed is exquisitely responsive to fluid status, temperature and hyper-active sympathetic tone as is often induced by post-operative pain.

Steroids and Chemotherapeutic Drugs

- Large doses or chronic usage of glucocorticoids reduce collagen synthesis and wound strength.
- Major effect is to inhibit the inflammatory phase of wound healing and the release of lysosomal enzymes
- Steroids also inhibit epithelialization and contraction and contribute to increased rates of wound infection, regardless of the time of administration
- All chemotherapeutic anti-metabolite drugs adversely affect wound healing by inhibiting early cell proliferation and wound DNA and protein synthesis

Metabolic Disorders

Diabetes Mellitus (DM)

- Uncontrolled DM results in reduced inflammation, angiogenesis and collagen synthesis.
- The accompanying large and small vessel disease contributes to local hypoxemia.
- Defects in granulocyte function, capillary ingrowth and fibroblast proliferation all have been described in Diabetes.
- Obesity, insulin resistance, hyperglycemia and diabetic renal failure contribute significantly and independently to the impaired wound healing observed in diabetics.
- Reduced expression of growth factors like VEGF, IGF 1, FGF 1, KGF and PDGF
- Diabetic fibroblasts and keratinocytes have diminished proliferation rates and collagen production.

Uremia

- It is associated with disordered wound healing.
- Experimentally, uremic animals demonstrate reduced wound collagen synthesis and breaking strength

Ionizing Radiation

- Ionizing radiation causes endothelial cell injury with endarteritis resulting in atrophy, fibrosis and delayed tissue repair

- Angiogenesis is *not* initiated
- Rapidly dividing cell populations like keratinocytes and fibroblasts are the most sensitive to radiation.

ABNORMAL WOUND HEALING

1. *Inadequate regeneration* e.g. CNS, corneal ulcers and non-union of bone.
2. *Inadequate scar formation* e.g. diabetic foot ulcers (DFU), pressure sores and stasis ulcers.
3. *Excessive regeneration* e.g. neuromas, hyperkeratosis in psoriasis and colonic polypi.
4. *Excessive fibrosis* e.g. hyper-trophic scars and keloids.

Surgical Infections

Definition

- Infections that should be treated by surgical intervention
- Infections following surgical procedure occurring at the wound site (surgical site infection [SSI] or at a distant site)

CLASSIFICATION

General Classification

According to duration

- *Acute infection* (<3 weeks)
- *Sub-acute infection* (3weeks - 2 months)
- *Chronic infection* (>2 months)

According to phase

- *Local phase*
- *Systemic phase*

Classifications of Soft Tissue Infections

The different types of soft tissue infections are summarized in Table 1.

Table 1: Soft tissue infections

Skin	Subcutaneous Tissue	Fascia
- Impetigo - Erysipelas - Lymphangitis - Folliculitis - Furuncle/carbuncle - Hidradenitis suppurativa - Pyoderma gangrenosum	- Wound infection - Synergistic necrotizing cellulites (Meleney's ulcer) - Non-clostridial gas-producing infection - Clostridial cellulites (gas gangrene)	- Necrotising fasciitis (Fournier gangrene) - Subfascial: - Non-clostridial myositis/myonecrosis - Streptococcal myositis - Clostridial myositis/myonecrosis

Classification of Hospital Acquired Infection (ICU)

- Sinusitis
- Neurological: meningitis - ventriculitis
- Chest: pneumonia - lung abscess - empyema
- Gastrointestinal: Enterocolitis
- Urinary tract infection (UTI)
- Catheter-related infections

ETIOLOGY

Protective Physiological Mechanisms

1. Intact epithelial surfaces
2. Chemical: Low gastric pH
3. Humoral: antibodies, complement and opsonins
4. Cellular: phagocytic cells, macrophages, polymorphonuclear cells and killer lymphocytes

Causes of Reduced Host Resistance

1. Metabolic malnutrition (including obesity), diabetes, uraemia, jaundice
2. Disseminated diseases: cancer and acquired immunodeficiency syndrome (AIDS)
3. Iatrogenic: radiotherapy, chemotherapy, steroids, surgical trauma.

Risk Factors of Wound Infection

1. Metabolic malnutrition (including obesity), diabetes mellites (DM), uraemia, jaundice
2. Disseminated diseases: cancer and AIDS
3. Iatrogenic: radiotherapy, chemotherapy, steroids, surgical trauma.
4. Immune suppression
5. Poor tissue perfusion
6. Foreign body
7. Poor surgical and disinfection techniques
8. Colonization and translocation in the gastrointestinal tract (GIT)

Organisms Involved in Wound Infection

- Staphylococcus - Streptococci
- Clostridia
- E-Coli
- Klebsiella - Psudomonas
- Fungi

SURGICAL WOUNDS

Classification of Wound Infection

The type of wound infections and related infection rate are listed in Table 2.

Table 2: Classification of Wound Infection

Type	Example	Infection Rate
- <i>Clean wounds</i>	Thyroidectomy	1-2%
- <i>Clean contaminated</i>	Cholecystectomy	2-5%
- <i>Contaminated</i>	Colon surgery	5-30%
- <i>Dirty</i>	Peritonitis	

Surgical Site Infection Prophylaxis

Pre-operative preparation

- Short hospital stay
- Control risk factors
- Bathing before surgery
- Shaving
- Bowel prep
- Antiseptic measures
- Antibiotics

Prophylactic measures

- Sterilization of equipment
- Shaving the surgical site
- Skin disinfection
- Sterile draping of operative field
- Scrubbing, gowning and gloving
- Irrigation of the wound with antibiotic solution
- Prophylactic systemic antibiotics peri-operatively

Clinical Picture

- It usually appears between the 5 and 10 days post-operatively
- *Symptoms:* fever, pain in the wound
- *Signs:* the wound (surgical site) becomes swollen, tender, red, hot, and fluctuant

Differential Diagnosis

Wound infection should be differentiated from other causes of *post-operative fever*, which include the following:

- Chest infection
- Deep vein thrombosis (DVT)
- Urinary tract infection (UTI)
- Seroma - Hematoma

SYSTEMIC INFECTIONS

Types

- *Bacteremia*: presence of pathogenic organisms in the blood stream without clinical picture of infection
- *Septicemia*: presence of toxic manifestation of infection
- *Pyemia*: pus in the blood (systemic or portal)

Classification

- *Sepsis*: SIRS with documented infection
- *Severe sepsis*: SIRS with evidence of one or more organ failure
- *MODS*: multiple organ dysfunction
- *MSOF*: multiple systemic organ failure

ACUTE INFECTIONS

Classification (Types)

Table 3 summarizes the different types of acute infections; non-specific or specific.

Table 3: Types of Acute Infections

Non-Specific Acute Infections	Specific Acute Infections
- Post-operative wound infection - Cellulitis - Erysipelas - Boil (furuncle) - Carbuncle - Hydradenitis suppurativa - Acute abscess - Acute lymphangitis and lymphadenitis - Bacteremia and septicemia	- Tetanus - Gas Gangrene - Necrotizing Fasciitis

A. Non-Specific Acute Infections

Cellulitis

Definition: It is an *invasive* non-suppurative infection of the loose connective tissue

Organisms: streptococci [common], staphylococci [occasionally], mixture

Clinical picture

- The affected area is red, indurated, hot and painful
- It spreads rapidly with ill-defined edge
- The skin may be the seat of blisters
- Fever
- Lymphangitis in the form of red streaks
- No suppuration
- In severe cases: patches of skin necrosis with sloughing of subcutaneous tissues

Differential Diagnosis

1. Contact allergy
2. Chemical inflammation
3. DVT

Treatment

1. Rest and elevation of the affected part
2. Antibiotic: penicillin

Erysipelas

Definition

- It is a rapidly spreading non-suppurative inflammation of the lymphatics of the skin caused by hemolytic streptococci.

Clinical Picture

- Systemically: Toxemia
- Locally: similar to cellulitis, but there are the following *differences*:
 1. The color of the skin is rose-pink
 2. The edge is well-defined
 3. There may be islets of inflammation beyond the spreading margin

Complications

- Facial erysipelas may lead to cavernous sinus thrombosis
- Septicemia
- Recurrent erysipelas may block the lymphatics leading to elephantiasis.

Treatment

- Isolation
- Similar to cellulitis

Boil (Furuncle)

- *Definition*: It is a *staphylococcal* infection of a hair follicle or a sebaceous gland.
- *Common sites*: Face, neck and axilla.
- *Clinical Picture*: It is common in patients with diabetes. It presents as a small painful indurated swelling, which is red, hot and very tender
- *Treatment*: Antibiotics and antiseptics.

Carbuncle

Definition

- It is infective gangrene of the subcutaneous tissues usually by *Staphylococcus aureus*.

Pathology

- Infection usually starts in a hair follicle
- It extends to the subcutaneous fat where other hair follicles get the infection.
- Multiple areas of necrosis and thrombosis of blood vessels occur.
- Patches of skin undergo sloughing and separate from the underlying granulation tissue

Clinical Picture

- It is common in immuno-compromised patients as in diabetics.
- The common sites: face, nape of the neck, and the back
- There is usually severe toxemia.
- It starts as a painful induration of the skin and subcutaneous tissues.
- The skin is red.
- The central part of the swelling becomes soft.
- Multiple areas of skin thin out and separate forming multiple sinuses.

Complications

- Local spread of infection.
- Pyemia and septicemia.
- Cavernous sinus thrombosis
- Epidural abscess or meningitis

Treatment

- Antibiotics (culture and sensitivity of the discharge).
- Control of diabetes.
- Surgical excision of sloughs

Hydradenitis Suppurativa

Definition

- Mixed staphylococcal and streptococcal infection of the *apocrine* sweat glands in the perineum or the axilla.

Clinical Picture

- It produces multiple abscesses and sinuses.

Treatment

- Surgical drainage of abscesses.
- Antiseptic and antifungal applications.
- Surgical excision of apocrine sweat-bearing skin followed by skin graft is essential.

Acute Abscess

Definition

- It is a localized suppurative inflammation.

Etiology

- Pyogenic organisms; the commonest are staphylococci (produce *coagulase* enzyme).

Pathogenesis

The organism reaches the tissues by:

- Direct access through wounds, scratches and abrasions.
- Local extension from an adjacent focus
- Lymphatic spread.
- Blood spread.

Pathology:

An abscess consists of three zones:

- 1- A *central* zone of coagulative necrosis
- 2- An *intermediate* zone of granulation tissue.
- 3- A *peripheral* zone of acute inflammation.

Sequel

- Resolution.
- Pointing and rupture.
- Spread infection locally, lymphatics or blood
- Chronicity.

Clinical Picture

- Locally: painful tender mass, skin is red, and edematous. Draining lymph nodes are usually enlarged and tender
- Systemically: fever, malaise, Headache, Tachycardia, Anorexia
- When pus forms:
 - 1- Fever becomes hectic.
 - 2- Skin shows pitting edema.
 - 3- Pain becomes throbbing.
 - 4- The inflammatory reaction becomes localized
 - 5- Fluctuation test becomes positive.
 - 6- There is shooting leucocytosis

Treatment

- Before suppuration: antibiotic, rest, hot fomentation and supportive general measures.
- After suppuration: surgical drainage. Culture and sensitivity and antibiotic (*if there are systemic manifestations*).

Acute Lymphangitis and Lymphadenitis

Acute lymphangitis is due to infection of lymph vessels by organisms usually *streptococci*.

Acute lymphadenitis is due to spread of infection along lymphatics from a septic focus in the drainage area to the lymph- nodes.

Treatment

- Antibiotics.
- Hot applications.
- Surgical drainage if suppuration occurs

B. Specific Acute Infections

Tetanus

Definition

- It is a specific anaerobic infection that is mediated by neurotoxin of *Clostridium tetani* and leads to nervous irritability and tetanic muscular contractions.

Etiology

- Organism: *Clostridium tetani* is *gram positive anaerobic bacillus* with a terminal spore giving the characteristic *drum-stick* appearance.
- Mode of Infection
 1. Wounds: -hypoxic, containing devitalized tissue or a foreign body.
 2. Umbilical stump: tetanus neonatorum

Pathogenesis

- The neuro-toxin is an *exotoxin* produced *locally* and reaches the central nervous system (CNS) along the blood stream, the motor nerves, or *both*.
- When the toxin reaches the nervous system, it is fixed by the motor cells and cannot be detected in the blood or cerebrospinal fluid (CSF).
- The anti-toxin can only neutralize the toxin before it gets fixed to the nervous tissue.
- The toxin increases the excitability of the motor cells of the medulla and spinal cord, so slightest stimuli produce violent spasm.
- Death results from exhaustion, hyperpyrexia, heart failure, asphyxia or pneumonia.

Clinical Picture

Incubation Period

- Symptoms during incubation period are vague such as tenderness, rigidity of the muscles, swelling at the site of wound, local twitches, restlessness, and anxiety.

Tonic Stage

- Pain and tingling in the area of injury.
- Limitation of movements of the jaw.
- Spasm of the facial muscles.
- Stiffness of the neck.
- Dysphagia.
- Laryngospasm.
- Hesitancy in micturition.

Clonic Stage

- Reflex paroxysms of violent muscular contraction.
- Relaxation is incomplete during the intervals between clonic contractions.
- Spasm of the intercostal muscles and diaphragm leads to long period of apnea.
- Temperature is elevated with profuse sweating.
- Marked tachycardia
- Laboratory finding: polymorph-nuclear leucocytosis

Prevention

- Immunization (active) every 7-10 years.
- Those previously immunized within 10 years: need a booster dose of tetanus toxoid.

- Those who received less than three doses: need a booster dose of tetanus toxoid and tetanus immunoglobulin (TIG) (passive).
- Individuals not previously immunized: need full immunization with tetanus toxoid and tetanus immunoglobulin

Treatment

- Neutralize toxin with TIG.
- Wound debridement.
- Avoid sudden stimuli.
- Muscle relaxant with mechanical ventilation may require tracheostomy.
- Aqueous penicillin G
- Nursing.

Gas Gangrene

Definition

- It is an acute spreading infection associated with gas formation and profound toxemia caused by *anaerobic spore-bearing bacilli* of the *clostridium* group.

Pathology

- Clostridia proliferate and produce toxins that diffuse into the surrounding tissue.
- The toxins destroy local circulation.
- This allows further invasion.
- Factors predisposing to gas gangrene:
 1. Lacerated wounds involving bulky muscles.
 2. Presence of foreign bodies or devitalized tissues.
 3. Ischemia of muscles.
 4. Infection by anaerobic bacteria

Bacteriology

- Organisms falls into two groups:
 1. Saccharolytic organisms: Cl. Welchii, Cl. Septicum, and Cl. edematiens
 2. Proteolytic organisms: Cl. Sporogenes, Cl. Histolyticum and Cl. tertium

Clinical Picture

- Incubation period varies from few hours to few days.
- Generally
 1. The patient is pale, anxious, and apprehensive.
 2. Temperature may be raised and there is marked tachycardia.
 3. Hands are cold and clammy.
 4. An icteric tinge may be present and there is oliguria.
 5. In severe case there is shock.
- Locally
 1. Pain and numbness in the affected area.
 2. Swelling and there may be crepitus with gas bubbles.
 3. A sanguineous discharge of a characteristic odor.
 4. The affected muscles brick red then greenish and finally *black* discoloration, do not contract, do not bleed if cut, the skin black.

Prevention

- Adequate debridement of wounds.
- Antibiotics.
- Avoid tissue hypoxia.

Treatment

- Wound management.
- Hyperbaric oxygenation.
- Antibiotics: penicillin

Necrotizing Fasciitis

Definition

- It is an invasive infection usually caused by a mixed microbial flora including microphilic streptococci, staphylococci, Gram-negative bacteria and anaerobes, especially streptococci, and bacteroids.

Pathogenesis

- The infectious process spreads along the fascial planes and results infectious thrombosis of the vessels passing between the skin and deep circulation.
- Superficial skin necrosis follows
- Hemorrhagic bullae appear as the first sign of skin death. Fascial and subcutaneous fat necrosis involves wider area than the skin

Clinical Picture

- There are manifestations of toxemia with fever and tachycardia.
- The skin shows hemorrhagic bullae and necrosis surrounded by oedema and inflammation.
- Crepitus is occasionally present.

Investigations

- Swab for culture and sensitivity
- At surgery: edematous, dull grey fascia and subcutaneous tissue with visible thrombi in penetrating vessels

Prevention

- Adequate debridement of wounds
- Antibiotics

Treatment

- Surgical
- Antibiotics
- Blood transfusion

PRINCIPLES OF ANTIBIOTIC MANAGEMENT

Cultures from:

- Exudate: The most reliable diagnosis for treatment. Both aerobic and anaerobic culture

- Blood: Diagnostic step for unknown source. Fail to capture causative organisms in bacteremia and unnecessary to diagnose sepsis
- Sputum
- Urine

Proper Choice of Antibiotics

- Bacteriostatic agents: *Prevent growth of bacteria*
- Bactericidal agents: *Actually kill bacteria*
- Effective agent against the infecting organism
- Adequate contact between agent and organism
- Absence of toxic side effect of the agent
- Augmentation of host defences to maximize antibacterial effects
- Administer antibiotics on empiric basis before the laboratory reports
- A combination of antibiotics for probable polymicrobial infection
- Culture and sensitivity test before antibiotic therapy
- Hazards
 1. Colonization
 2. The quantitative appearance of changes in the microflora that are induced by antibiotic therapy
 3. Super-infection
 4. A new microbial disease introduced or potentiated by antibiotic therapy
 5. *Super-infection* is frequently the result of *colonization*.
- For potentially contaminated wounds
- Only an adjunct and *not* a substitute to good surgical technique

Type of Wound

- *Clean procedure*: No antibiotics are necessary
- *Clean contaminated procedure*: Contact of the interior of respiratory, urinary, GIT
- *Contaminated procedure*: Complicated by gross spillage of intestinal contents or wounds secondary to trauma
- *Dirty wounds*: In contact with intra-abdominal or peri-rectal abscess

Other Principles

- Early and enough for adequate tissue and body fluid levels
- Being necessary to maintain adequate tissue levels intra-operatively
- Length of operation and serum half-life of antibiotics

Infection and the Surgeon

- Full face mask
- Water-proof gowns
- Boot
- Double gloving
- Only essential personnel in the operating room
- No unnecessary movement
- Slow, meticulous, operative techniques

Principles of Surgical Oncology

PRINCIPLES OF SURGICAL ONCOLOGY

Definition

Surgical oncology is the use of surgery in diagnosis and treatment of cancer. Surgery can be used by itself to treat the cancer or it can be done with other methods.

Role of Surgery

- Role of Surgery is to diagnose cancer, stage cancer, provide radical cure or plan other treatment techniques
- Not all cancers can be treated with surgery. The tumor may not be safely removed without causing damage for the patient.
- Surgery in cancer is used to remove the whole cancer (radical curative), remove some, but not all the tumor (debulking or palliative). A surgeon may not be able to safely remove all of the cancer because it sometimes invades other important organs.
- Another role of surgery is to reform the body's condition after a surgery. This is called *reconstructive* or restorative surgery. Examples are breast reconstruction or reversal of colostomy.

Role of Surgery with Other Therapies

- Surgery may be the only treatment for some patients. For many people, surgery is used along with other treatments. These can be chemotherapy, biological, targeted or immune therapies, radiation therapy, and hormone therapy. Other surgeries, such as those used in interventional specialties, may be used too.
- Neoadjuvant therapy: This means the surgery is done after some other form of treatment.
- Adjuvant therapy: This means the surgery is done before the other form of treatments.
- Despite the advances in medical and radiation oncology, surgery is still the only modality with the potential to cure most solid cancers. Surgeons have a pivotal role in cancer treatments and research, leading the diagnostic and treatment pathways for most cancers from counselling patients about their diagnosis through to surgery and aftercare. They have also led many of the great advances in cancer research.
- However, cancer care has evolved very rapidly over the last few decades and therefore a new type of surgeon is needed to keep pace with these changes. No longer is surgery alone the only treatment for most solid malignancies but a combination of surgery and

multi-modal therapies (with highly focused radiotherapy, targeted molecular therapies and poly-chemotherapy) becoming the modern standard of care.

- As a result, the surgeon, can no longer work in isolation but must lead a multidisciplinary team. The surgeon must be more than just a technician and must understand the biology and natural history of the disease as well as the contributions made by other disciplines to the cancer patients' treatment. It is at this point that the surgeon becomes a surgical oncologist. Examples include: collaboration with radiation and medical oncologists on the use of neoadjuvant chemotherapy or radiotherapy to enhance or permit surgery possible and on the indications for adjuvant therapy after surgery. Collaboration with radiologists to plan surgery or optimize resection margins.

Disadvantages of Surgery

1. Pain
2. Bleeding
3. Infection
4. Delayed wound healing. Bleeding.

NEOPLASMS

Definition

Cell population characterized by:

- Excessive, non-useful proliferation
- Unresponsive to normal control mechanisms
- Harmful to adjacent tissue.

Histological Cascade

- Normal cells
- Hyperplasia
- Atypia
- Dysplasia
- Carcinoma in-situ
- Invasive malignancy

Tumour Grading

- **G1:** Well differentiated
- **G2:** Moderated differentiated
- **G3:** Poorly differentiated
- **G4:** Undifferentiated

Types of Neoplasms

- Benign
- Locally-malignant
- Malignant: Carcinoma (95%), or sarcoma (5%)

Signs of Changing from Benign into Malignant

- Rapid growth
- Bleeding
- New vasculature
- Hard consistency
- Fixation
- Infiltrating nearby structure
- Dissemination: lymph nodes or remote secondaries

Characteristics of Sarcoma *versus* Carcinoma

- Younger age
- Rapid rate of growth
- Rapid local invasion
- More blood-born metastasis
- Worse prognosis
- Less sensitive to radiotherapy and chemotherapy

Premalignant Lesions

- Barrett's esophagus: cancer esophagus
- Bowen disease of skin: squamous cell carcinoma (SCC) (epithelioma)
- Ulcerative colitis: cancer of the colon
- Chronic atrophic gastritis: gastric cancer
- Familial intestinal polyposis: cancer of the colon
- Xeroderma pigmentosa: epithelioma

Techniques for Obtaining Tissue Biopsy

- Needle biopsy: fine-needle aspiration cytology (FNAC), Core tissue biopsy
- Incisional biopsy
- Excisional biopsy
- Frozen-section biopsy
- Brush cytology

Most Common Sites of Metastasis

- Bone
- Brain
- Lung
- Liver

Routes of Spread

1. Direct extension: to nearby organs
2. Lymphatic spread: common in epithelial neoplasms (carcinoma)
3. Vascular spread: more common in sarcomas
4. Spread through serous cavities: peritoneal seeding (cancer of the stomach)

Clinical Staging

- Stage I: cancer confined to its primary site
- Stage II: spread to regional lymph nodes (LNs)
- Stage III: more locoregional advanced disease
- Stage IV: metastasis to distant sites

Prognosis

- Best 5-year survival in patients with cancer of *skin, cervix, uterus and bladder*; while lowest survival is with *pancreatic and gastric cancer*
- Females tend to have a higher 5-year survival than males.

Principles of Cancer Surgery

- Complete excision of tumor mass with nearby extension and LN involvement (*radical operation*)
- *Prevention of tumor cell implantation during surgery*
- Surgery for palliation: to improve the quality of life e.g. relief of intestinal obstruction, removal of mass causing pain

Follow-up

- Response to treatment
- Relapse
- Diagnosis of new tumour

Psychological Aspects

- On diagnosis: shock, denial, sadness
- On treatment: depression, loneliness
- On follow up: worry, emotionless

Groin Hernias

HERNIA: INTRODUCTION

Definitions

- Hernia means “*To bud*’ or *‘to protrude*’, *‘off shoot*’ (Greek) - *‘rupture*’ (Latin).
- A hernia is defined as "an area of weakness or disruption of the fibro-muscular tissues of the body wall".
- Often hernia is also defined as "an actual anatomical weakness or defect".
- Hernia is also defined as "an abnormal protrusion of a viscous or a part of a viscous through an opening, artificial or natural with a sac covering it".

Incidence

- About 15% of males AND 5% of females will develop groin hernia.
- Approximately, 75-85% of abdominal wall hernias are groin hernias (inguinal and femoral)
- *Inguinal hernia is the most common hernia (73%)* because of weakness of the muscular anatomy in the inguinal region and presence of natural weakness like the deep inguinal ring and cord structures. *Incisional hernia* is next to inguinal hernia in occurrence. Incidence of femoral hernia is 17%; umbilical 8.5% and others 1.5% (excluding incisional hernia).

Common and Rare Sites of Abdominal Wall Hernia (Figure 1)

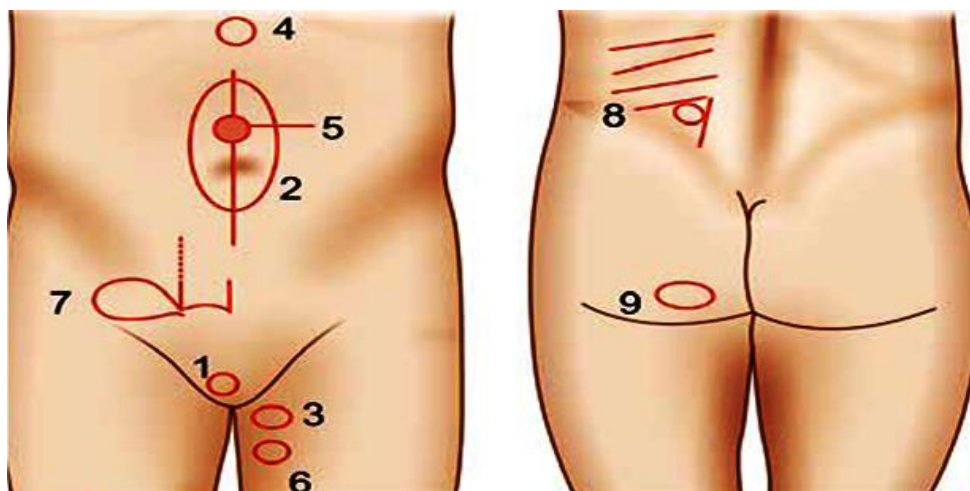


Figure 1. Common sites of abdominal wall hernia (1. Inguinal, 2. Incisional, 3. Femoral, 4. Epigastric, 5. Umbilical). Rare sites (6. Obturator, 7. Spigelian, 8. Lumbar, 9. Gluteal)

Etiology

- Straining
 - o Lifting of heavy weights.
 - o Chronic cough (tuberculosis, chronic bronchitis, bronchial asthma, emphysema).
 - o Chronic constipation (habitual, rectal stricture).
 - o Urinary causes: Benign prostatic hyperplasia (BPH), carcinoma prostate (old age), stricture urethra (young age), and phimosis and meatal stenosis (very young age).
- Obesity.
- Pregnancy and weakness of pelvic musculature (especially femoral hernia in females).
- Smoking.
- Ascites.
- Appendectomy through McBurney's incision may injure the ilio-inguinal nerve causing right sided *direct* inguinal hernia (DIH).
- An indirect inguinal hernia (IIH) occurs in a congenital, preformed sac (remains of processus vaginalis). Chance of presence of bilateral preformed sac is 60%.
- Familial collagen disorder - *Prune Belly syndrome*.
- Acquired herniation is also probably due to collagen deficiency called *metastatic emphysema of Read*.

Parts of Hernia

1. Sac

- It is a diverticulum of peritoneum with mouth, neck, body and fundus.
- The "neck" is *narrow in indirect sac but wide in direct sac*.
- Body of the sac is thin in infants, children & in indirect sac but is thick in direct & long-standing hernia.
- Hernia *without* neck: Those hernias with larger mouth lack neck, e.g. direct hernia, incisional hernia.
- Hernia *without* sac: Fatty hernia of linea alba = protrusion of extra-peritoneal pad of fat.

2. Coverings

- Coverings of the sac are the layers of the abdominal wall through which the sac passes.

3. Contents of Sac

- Omentum = (*Omentocele*). Initially it can be reduced easily, but difficult later.
- Intestine = *Enterocoele* (Commonly small bowel, but sometimes large bowel). It is difficult to reduce the contents initially.
- A portion of circumference of bowel = ***Richter's hernia***.
- UB may be the content or part of the posterior wall of the sac = *cystocoele*.
- Ovary, often with Fallopian tube.
- Meckel's diverticulum = ***Littre's hernia***.
- *Fluid*: Fluid is secreted from congested bowel or omentum. It may be an infected fluid or ascitic fluid or blood from the strangulated sac.

Classification of Hernia

Classification I: Clinical

1. *Reducible Hernia*: Hernia gets reduced on its own or by the patient or by the surgeon. Intestine reduces with gurgling and is difficult to reduce the first portion. Omentum is doughy and it is difficult to reduce the last portion. Expansile impulse on cough is present.
2. *Irreducible Hernia*: Here contents cannot be returned to the abdomen due to narrow neck, adhesions, over-crowding. Irreducibility predisposes to strangulation.
3. *Obstructed Hernia*: It is an irreducible hernia with obstruction, but blood supply to the bowel is not interfered. It eventually leads to strangulation.
4. *Inflamed Hernia* It is due to inflammation of the contents of the sac, e.g. appendicitis, salpingitis. Here, the hernia is tender but not tense; overlying skin is red and edematous.
5. *Strangulated Hernia*: It is an irreversible hernia with obstruction to blood flow. The swelling is tense, tender, with no impulse on cough and with features of intestinal obstruction, which may be absent in case of omentocele, Richter's or Littre's hernia.

Classification II: Etiological

1. *Congenital (Common)*: It occurs in a preformed sac/defect. Clinically may present at a later period due to any of the precipitating causes like in indirect inguinal hernia (IIH).
2. *Acquired*: It is 2nd to any causes which raise the intra-abdominal pressure leading into weakening of the area like in direct inguinal hernia (DIH).

Classification III: According to Contents

1. Omentocele (omentum)
2. Enterocoele (intestine)
3. Cystocoele (Urinary bladder).
4. Littre's hernia—Meckel's diverticulum (*NB*: Littre described Meckel's diverticulum in a hernial sac 81 years before Meckel was born).
5. Maydl's hernia.
6. Sliding hernia.
7. Richter's hernia—part of the bowel wall.

Classification IV: According to Site

1. Inguinal hernia (in the inguinal canal).
2. Femoral hernia (in the femoral canal).
3. Obturator hernia.
4. Diaphragmatic hernia.
5. Lumbar hernia.
6. Spigelian hernia.
7. Umbilical hernia.
8. Epigastric hernia.

INGUINAL HERNIA

Surgical Anatomy of Inguinal Canal

- **Superficial inguinal Ring (SIR):** It is a triangular opening in the external oblique aponeurosis (EOA) and is 1.25 cm *above* the pubic tubercle. The ring is bounded by a supero-medial & infero-lateral crus. Normally, the ring does *not* admit the tip of little finger.
- **Deep Inguinal Ring (DIR):** It is a U-shaped condensation of the transversalis fascia, lies 1.25 cm *above* the inguinal ligament *midway* between the symphysis pubis & anterior superior iliac spine (ASIS).
- **Inguinal (Poupart's) Ligament:** It is formed by the lower border of the EOA, which is thickened and folded backwards on itself, extending from the ASIS to the pubic tubercle.
- **Inguinal Canal:** It is an oblique passage in lower part of abdominal wall, 4 cm long, situated above the medial ½ of inguinal ligament, extending from the DIR to the SIR. *In infants*, both rings are *superimposed* without obliquity of the inguinal canal. *In females*, the inguinal canal is called the "**canal of Nuck**".
- **Coverings of spermatic cord**
 1. Internal spermatic fascia from the fascia transversalis
 2. Cremasteric fascia.
 3. External spermatic fascia from EOA is seen below the SIR, in the scrotum.
- **Boundaries of inguinal canal** (Figure 2)
 1. *Anteriorly:* EOA
 2. *Laterally:* conjoint muscle (Arched fibers of internal oblique).
 3. *Posteriorly:* Inferior epigastric artery (IEA), fascia transversalis & conjoined tendon *medially*.
 4. *Above:* Conjoined muscle
- *Below:* Inguinal ligament.

Contents of inguinal canal

- ❖ Spermatic cord in males
- ❖ Round ligament in females
- ❖ Ilioinguinal nerve

Contents of spermatic cord

- ❖ Vas deferens
- ❖ Artery to vas
- ❖ Testicular and cremasteric artery
- ❖ Genital branch of genitofemoral nerve
- ❖ Pampiniform plexus of veins
- ❖ Remains of processus vaginalis
- ❖ Sympathetic plexus around the artery to vas

Defence mechanism of inguinal canal

- ❖ Obliquity of inguinal canal
- ❖ Arching of conjoint tendon
- ❖ 'Shutter mechanism' of internal oblique
- ❖ 'Ball valve mechanism' due to contraction of cremaster muscle which plugs to superficial ring
- ❖ When external oblique muscle contracts, intercrural fibres of superficial ring appose causing 'slit valve mechanism'
- ❖ Hormones

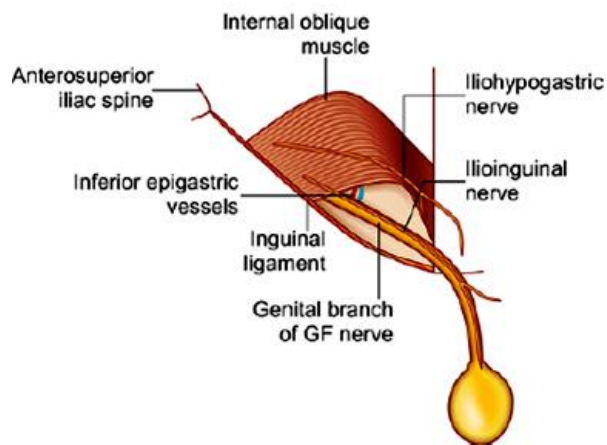


Figure 2. Boundaries (+ contents) of inguinal canal

Classification of Inguinal Hernia

Classification I

Anatomical Classification of Inguinal Hernia (Figure 3)

1. **Indirect inguinal hernia (IIH):** It comes out through the DIR (deep internal ring) along with the cord. It lies *lateral* to the IEA.
2. **Direct inguinal hernia (DIH):** It occurs through the posterior wall of the inguinal canal through 'Hesselbach's triangle' (bounded *medially* by the lateral border of rectus muscle, *laterally* by the IEA & *inferiorly*, by the inguinal ligament).

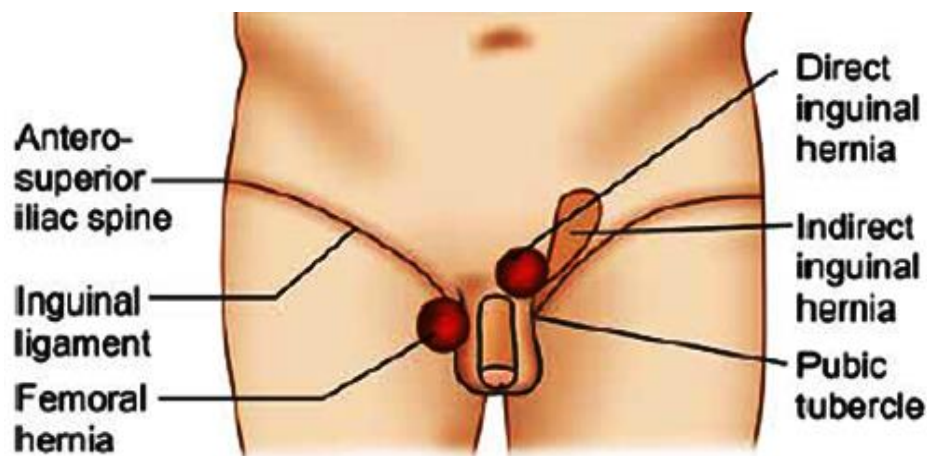


Figure 3. Anatomical location of IIH and DIR (and femoral hernia)

Classification II

According to the Extent

1. Incomplete

- *Bubonocoele:* The sac is confined to the inguinal canal.
- *Funicular:* The sac crosses SIR, but does not reach the bottom of the scrotum.

2. Complete:

- The sac descends to bottom of the scrotum

N.B. *Saddle-bag or pantaloons hernia* has both IIH and DIH

Newer Classifications of Hernia

I. Gilbert Classification

Type I	- Hernia has got snug the DIR through which a peritoneal sac passes out as indirect sac.
Type II	- Hernia has a moderately enlarged DIR, which admits one finger but < 2-finger breadth. Once reduced, it protrude during coughing or straining
Type III	- Hernia has got large DIR with a defect >2-finger breadth. Hernia descends into the scrotum or with sliding hernia. Once reduced, it immediately protrudes out <i>without</i> any straining
Type IV	- It is <i>direct hernia</i> with large full blow out of the posterior wall of the inguinal canal. The DIR is intact
Type V	- It is a <i>direct hernia</i> protruding out through punched out hole/defect in the Transversalis fascia. The DIR is intact
Type VI	- <i>Pantaloon / double hernia</i>
Type VII	- <i>Femoral hernia</i>

N.B. Types VI & VII are **Robbin's Modification**.

II. Nyhus Classification

Type I	- Indirect inguinal hernia (IIH) with normal DIR.
Type II	- IIH with dilated DIR
Type III	- Posterior wall defect a. Direct inguinal hernia (DIH). b. Pantaloon hernia. c. Femoral hernia
Type IV	- Recurrent hernia

A. INDIRECT (OBLIQUE) INGUINAL HERNIA (IIH)

Incidence

- *General incidence:* It is the commonest type of hernia (65%) in males.
- *Age:* It is more common in younger age group as compared to DIH, which is more common in the elderly.
- *Side:* It is more common on right side in the first decade of life, but incidence is equal on both sides in the 2nd decade. It is bilateral in 30% of cases.

Pathology

- *The sac* is thin in IIH. The neck is narrow and lies *lateral* to the inferior epigastric vessels.
- *Coverings* of IIH from inside out:
 1. Extra-peritoneal tissue
 2. Internal spermatic fascia
 3. Cremasteric muscle
 4. External spermatic fascia
 5. Skin.

Types (Figure 4)

1. *Bubonocoele*

The hernia in bubonocoele is limited to the inguinal canal.

2. *Funicular*

The processus vaginalis is closed just above the epididymis. Contents of the sac can be felt separately from testis, which lies below the hernia.

3. *Complete (Scrotal)*

The testis appears to lie *within* the lower part of hernia. It can occur in any age group. It occurs in a congenital preformed sac (processus vaginalis). More commonly, contents descend into the pre-existing sac only when there are precipitating causes, which force the content down.

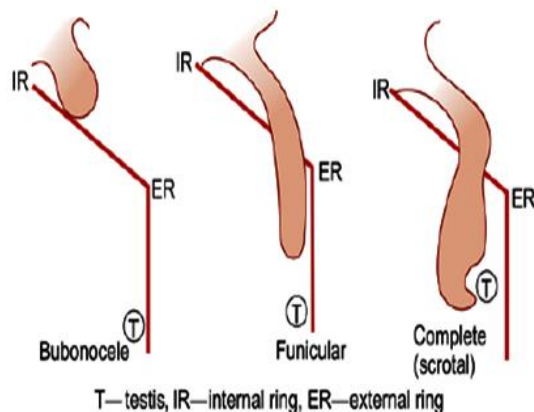


Figure 4. Types of indirect inguinal hernia (IIH): bubonocoele – funicular – complete (scrotal)

Clinical Features

- **Gender:** It is more common in males (M:F ratio = 20:1).
- Patient presents with dragging pain and swelling in the groin, which is better seen while coughing and standing; and felt together with an *expansile impulse*.
- The swelling is confined to the inguinal region in bubonocoele (Figure 5), reaches down to the middle of the scrotum in the funicular type (Figure 6), and descends down to the scrotum completely in the scrotal (complete) type, which may reach the thigh (*giant*) (Figure 7).



Figure 5. Left bubonocoele



Figure 6. Left Funicular IIH



Figure 7. Complete

- *Contents* are small bowel, large bowel, omentum or combination of all these. Sometimes, in females, the ovary and tubes may be the content. In infants, the swelling appears when the child cries and is often translucent. *Differences between enterocele and omentocele are:*

Enterocele	Omentocele (Epiptocele)
- The first part is difficult to reduce but the last part is easier. - Gurgling sound on reduction of contents. - Peristalsis may be seen (inspection). - Resonant on percussion. - Bowel sounds may be heard	- The first part is easier to reduce but the last part is difficult. - It has a doughy feeling - No peristalsis seen - Dull on percussion - Bowel sounds not heard

- It is usually *reducible*, but can be *complicated* with irreducibility, inflammation, obstruction or strangulation.
- **Internal ring (DIR) occlusion test**
The DIR ring is located half an inch above the mid-inguinal point (center point between ASIS and symphysis pubis). After reducing the contents, in lying down position, the internal ring is occluded using the thumb. The patient is asked to stand up and cough. If a swelling appears medial to the thumb, then it is a DIH (Figure 8). If the swelling does *not* appear and on releasing the thumb, it appears during coughing, then it is an IIH.
- **Ring invagination test (obsolete):** After reduction of contents, the little finger/index finger of the examiner is invaginated from the bottom of the scrotum, pushed up & rotated to enter the SIR. The impulse on coughing is felt at the tip of the invaginated finger.
- **Zieman's test:** The examiner places his index finger on the DIR & the middle finger on the SIR and the ring finger over the saphenous opening. The patient is asked to cough or to hold the nose and blow. If the impulse is felt on the index finger, it is an IIH.
- **Head or leg rising test:** To look for abdominal wall muscle tone and Malgaigne bulgings.
- **Valsalva maneuver:** It is also used to check the tone of abdominal wall muscles.
- **General examination:** Abdominal, chest and urological examination is done to look for any precipitating factor such as chronic bronchitis, bronchial asthma, ascites, stricture urethra, or benign prostatic hyperplasia. **PR** examination is a must.

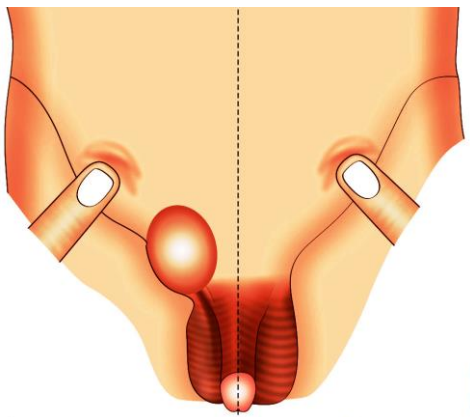


Figure 8. DIR test: Appearance of the swelling = DIH

Use five fingers of the hand to complete all tests for hernia
❖ Thumb—for deep ring occlusion test ❖ Index, middle and ring fingers for Zieman's test ❖ Little finger for superficial ring invagination test
Rules of hernia examination
❖ Never forget to examine opposite side ❖ Never forget to do per rectal examination ❖ Never forget to examine urethra ❖ Never forget to check abdominal muscle tone

Differential diagnosis
❖ Hydrocele—infantile/encysted/large vaginal/bilocular ❖ Undescended testis ❖ Femoral hernia ❖ Lipoma of the cord ❖ Hydrocele of the canal of nuck (in females) ❖ Inguinal lymph node enlargement ❖ Groin abscess

- **Differences between IIH and DIH**

Criteria	DIH	IIH
<i>Incidence</i>	Less common	More common
- Occurrence	Old	Any age (young & middle age)
- Age	More likely to be bilateral	Usually unilateral
- Side		
<i>Etiology</i>	Usually acquired weakness	Mainly congenital defect
<i>Pathology</i>		
- Sac	Posterior to the cord	Within the layers of the cord
- Neck of the sac	Medial to the IEA, wide.	Lateral to the IEA, narrow
- Coverings	EOA.	Cord layers + EOA
- Parietal defect	Weak fascia transversalis	DIR
<i>Clinical Picture</i>		
- Shape	Hemispherical	Oblong
- Descent to scrotum	Negative or very rare	Common
- Reduction	Backwards	Upward, backward & laterally
- Descent	Forwards	Downward, forward & medially
- SIR size	Normal	Wide
- SIR test	Impulse felt on finger pulp	Impulse felt on finger tip
- DIR test	Negative	Positive
<i>Complications</i>	Less common due to wide neck	More common (narrow neck)
<i>Surgery</i>	Herniotomy: usually unnecessary	Usually necessary (long sac)

Investigations

- Laboratory investigations: CBC - coagulation profile -
- Chest X-ray to rule out chronic bronchitis.
- Ultrasound (US) of the abdomen.
- Tests relevant to the precipitating causes.

Treatment

- *Treatment is always surgery. Surgical repair reconstructs the inguinal canal as it is physiologically, with 2 rings, one abdominal, the other subcutaneous; and with 2 walls, one posterior and the other anterior, between which the spermatic cord passes obliquely.*

- Precipitating causes should be treated first, like TURP for BPH, dilatation of stricture urethra, treatment of chronic bronchitis. Patient is advised to avoid smoking.
- *In infants*: Whether it is hernia or hydrocele, only *herniotomy* is done through inguinal approach (*Michaelis plank operation*).
- *In adults*: It includes **herniotomy** (excision of hernial sac) and **herniorrhaphy** or **hernioplasty (ideal)** (strengthening of the posterior wall of inguinal canal either by repair or mesh, respectively). Hernioplasty is the present choice (ideal) for all groin hernias. A mesh is placed either onlay/underlay (over the conjoint tendon to inguinal ligament) or inlay (in the pre-peritoneal space). Polypropylene mesh is used. Herniotomy is done prior to mesh placement. TEP (*Totally extra-peritoneal laparoscopic pre-peritoneal mesh repair*) is the preferred method. (For detail refer hernioplasty in later part of this chapter).

Herniotomy

- **Anesthesia**: Spinal, general or local anesthesia.
- **Incision**: 1.25 cm above and parallel to the medial 2/3 of the inguinal ligament
- **Procedure**
 - o The 2 layers of superficial fascia (outer *Camper's* fascia, inner *Scarpa's* fascia) are incised.
 - o The EOA is incised. The upper leaf is reflected above and the lower leaf is reflected downwards to expose the inguinal ligament. The ilio-inguinal nerve is safeguarded.
 - o The cremasteric muscle is opened. Cord structures are dissected. The sac, which is pearly white in color and lies anterior and lateral to the spermatic cord is identified.
 - o Dissection is usually started from the fundus and extended towards the neck, which is identified by extra-peritoneal fat. The neck is narrow and is lateral to the IEA.
 - o The sac is opened at the fundus, contents, if any, are reduced and the sac is twisted so as to prevent the content from coming back. It is transfixed using absorbable suture material (Vicryl 2-0) and is excised distally.

Herniorrhaphy

Modified Edwardo Bassini's (1887) Herniorrhaphy (Figure 9)

- It is strengthening of the posterior wall of the inguinal canal by approximation of the conjoint tendon and inguinal ligament using interrupted non-absorbable mono-filament sutures {polypropylene}. The medial most stitch is taken from the periosteum of pubic tubercle (*called as key or Bassini's stitch*). The EOA and other layers are closed.
- Multifilament suture material like silk is *not* used as it may precipitate infection & the tensile strength is not as good as monofilament suture material.
- Continuous suture is *not* used as it compromises the blood supply and interferes with proper healing; and strength will not be as adequate as interrupted sutures.
- Earlier, the commonest surgery done for inguinal hernia was modified Bassini's repair. Now hernioplasty is the commonly done procedure for both DIH and IIH. In direct hernia, the sac is usually *not* opened in contrast to indirect hernia.

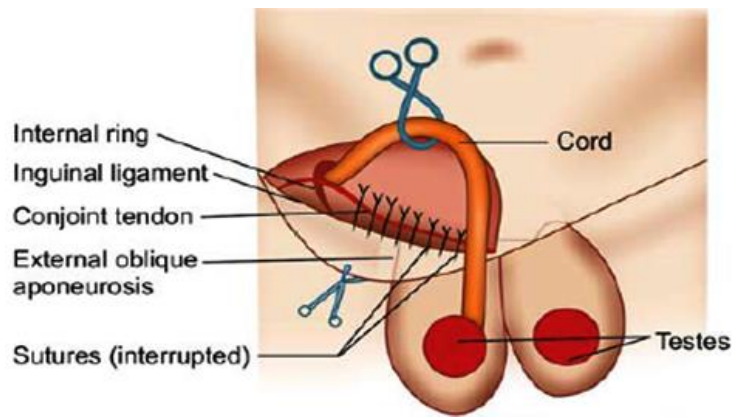


Figure 9. *Modified Bassini's repair:* Approximation of the conjoint tendon to the inguinal ligament using interrupted non-absorbable mono-filament sutures.

Lytle's Repair

- Often, the internal ring (DIR) is narrowed by placing interrupted sutures over the medial side of the ring to the transversalis fascia using interrupted Vicryl sutures in order to narrow the ring and push the cord laterally (Figure 10)

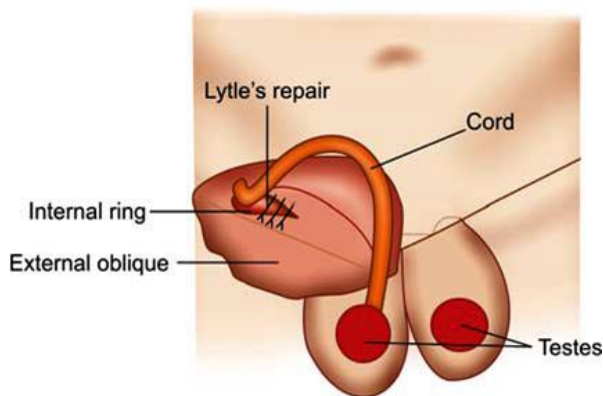


Figure 10. *Lytle repair:* Narrowing of the internal ring (DIR) using interrupted sutures

Shouldice Repair (Toronto)

- The transversalis fascia is thin, but tough and so double-breasting of this fascia using continuous non-absorbable sutures strengthens the posterior wall of the inguinal wall.
- It is a multi-layered repair. After herniotomy, the transversalis fascia is incised along the line of the wound from the DIR to the pubic tubercle.
- The lower flap of fascia is sutured to posterior part of the upper flap. Upper flap is sutured to the inguinal ligament. It causes double breasting of the transversalis fascia. Then conjoint tendon and inguinal ligament is further approximated by 2 layers of continuous sutures.
- The EOA is sutured in 2 layers (double-breasting) *in front of* the cord. Hence the original Shouldice repair is a *6-layered procedure*; first 2 layers of transversalis fascia, next 2 layers of conjoint tendon and last 2 layers of EOA. Suture material used here is fine steel wire of 34 gauge (in original Shouldice repair) or polypropylene or polyethylene.
- Recurrence rate (RR) is 1%.

Berliner Modified Shouldice Repair

- It involves double-breasting of the transversalis fascia like in Shouldice repair and a single- layer closure of the EOA *without* any additional two-layered repair of the conjoint tendon to the inguinal ligament.

McVay Operation—1940 (*Cooper's Ligament Repair*)

- It is done by placing interrupted sutures between the transversalis fascia to Cooper's ligament (superior pubic ligament) starting from the pubic tubercle medially towards the femoral sheath and later continued as suture repair between the transversalis fascia and iliopubic tract laterally up until entrance of the cord is reached.
- It is a pure tissue repair. It requires relaxing vertical incision at the lateral border of the anterior rectus sheath, from the pubic tubercle extending superiorly for 4 cm.
- It covers all three groin defects; DIR, IIH and femoral hernia.

Hernioplasty

- A *prosthetic mesh* (e.g. Prolene, Vipro, ultrapro) is used to strengthen the posterior wall of the inguinal canal.
- *Onlay*: e.g. **Lichtenstein** repair (Figure 11).
- *Underlay*: e.g. PHS repair (**Gilbert's**), **Stoppas**, totally extra-peritoneal pre-peritoneal (**TEP**), trans-abdominal pre-peritoneal (**TAPP**) mesh repair.
- RR is significantly reduced by prosthetic mesh repair. But incidence of mesh *inguinodynia* increases due to entrapment of ilio-inguinal or ilio-hypogastric nerves in the mesh. Even though, it is still the *commonly* done repair for inguinal hernia.
- Not used in children, in strangulated hernia or sepsis; only tissue repair is done.

Complications of herniorrhaphy	
❖	Haemorrhage
❖	Haematoma, seroma
❖	Infection—1-5%
❖	Haematocele
❖	Post-herniorrhaphy hydrocele, lymphocele
❖	Hyperaesthesia over the medial side of inguinal canal due to injury to iliohypogastric nerve—neuralgia (15%)
❖	Recurrence—10-15%
❖	Osteitis pubis
❖	Injury to urinary bladder/bowel
❖	Testicular atrophy, penile oedema rarely can occur

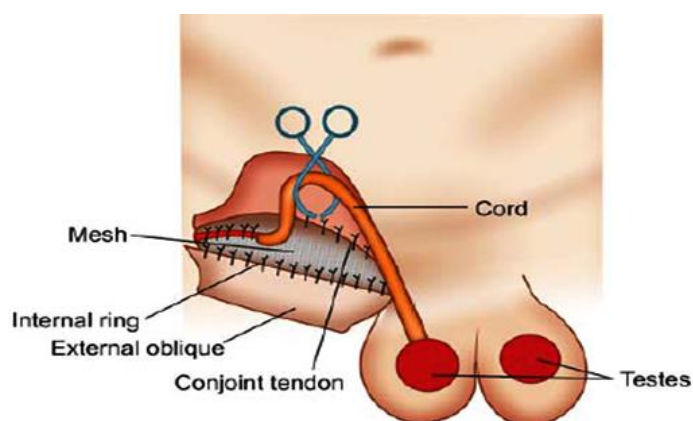


Figure 11. Hernioplasty *Lichtenstein repair*: The mesh is placed between the conjoint tendon & the inguinal ligament (*tension-free*)

2. DIRECT INGUINAL HERNIA (DIH)

Incidence

- 10-15% of the hernias are direct and 35% of inguinal hernias are direct.
- 50% of direct hernias occur bilateral.
- It is uncommon in females and children.

Surgical Anatomy

- It is always *acquired*, due to weakening of posterior wall of inguinal canal.
- Hernia is *medial* to the IEA with a wide neck.
- The sac is thick & often the medial wall or content may be the urinary bladder (UB).
- *Coverings* of DIH from inside out: (1) extra-peritoneal tissue, (2) internal spermatic fascia, (3) cremasteric muscle, (4) external spermatic fascia & (5) skin.
- Direct hernia occurs through **Hesselbach's triangle**, which is bounded by the inferior epigastric artery (IEA) *laterally*, lateral border of the rectus abdominis *medially* and the inguinal ligament *below*. It is divided into medial and lateral halves by the obliterated umbilical artery (*medial umbilical ligament*) (Figure 12). Thus, DIH is classified as *medial* or *lateral* depending on which part of the Hesselbach's triangle, is arising from.

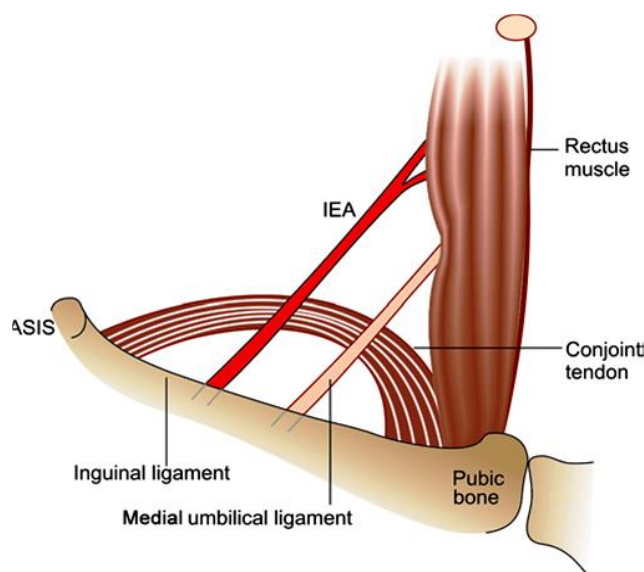


Figure 12. Hesselbach's triangle: It is bounded by the IEA laterally, lateral border of rectus medially and inguinal ligament (medial half) below.

Clinical Examination

- *Internal ring test* (refer back)
- *Malgaigne bulgings* are often seen in these patients on clinical examination more than in IIH. They are protrusions of abdominal wall muscle during leg-raising test as weak, soft, supple, swellings, which signify poor abdominal muscle tone.
- The DIH rarely descends into the *scrotum* and *strangulation* is *not* as common as in indirect IIH.
- However, in long-standing cases, it can descend down to the scrotum and strangulation can occur.

Treatment

Surgical Repair

- Usually the direct sac is *not* opened.
- Care should be taken at the medial aspect due to the presence of UB (*the UB should be emptied* before surgery).
- Ideally, *hernioplasty (mesh repair)* is done.
- In case of bilateral hernia, mesh repair can be done on both sides together.
- Laparoscopic approach or supra-pubic approach may be better in bilateral cases.

Funicular Direct Hernia (Pre-vesical Hernia) = Ogilvie's hernia

- It is a type of DIH, which is prone for strangulation.
- It is a narrow-necked hernia with pre-vesical fat and portion of the urinary bladder and/or intestine that herniate through a small defect in the medial part of the conjoined tendon just above the pubic tubercle.
- It occurs in elderly males.

3. RECURRENT INGUINAL HERNIA

Incidence is 10%. Recurrence within 3 years it is called “early”; after 3 years “late”.

Predisposing Factors

Pre-operative	Operative	Post-operative
- Smoking. - Straining (e.g. chronic cough, chronic constipation or difficult micturition). - ↑ intra-abdominal pressure (straining, tumor, ascites) - General factors (e.g. old age, anemia, and hypo-proteinemia).	- Tension on the sutures. - Inadequate technique (e.g. improper suture material used) - Weakness of the anterior abdominal wall (AAW)	- Infection (50%). - Hematoma formation. - Retained sac in pantaloon hernia. - Post-operative straining.

Recurrence Rate

- Bassini's repair (10%), Shouldice repair (1%), Hernioplasty (1-3%), other types (5%).
- *True or false recurrences?* If hernia occurs in the inguinal region after inguinal hernia repair it is called *true recurrence*, but if another groin hernia (e.g. femoral) appears, it is called *false recurrence*. Currently, any groin hernias that reappears, it is a *true recurrence*.

Clinical Features

- Similar to inguinal hernia. Defect is usually narrow and more likely to strangulate.
- It can be *medial* or *lateral recurrence* depending on the location of the sac.
- Medial recurrence is common (tension on suture line is greatest near pubic tubercle).

Treatment

- Treatment of the cause of recurrence is followed by *hernioplasty*.
- **Laparoscopic (TEP/TAPP) approach** is better for recurrent hernia.
- In **open repair**, *pre-peritoneal mesh repair* is ideal.
- In elderly people, **Koontz orchidectomy** or cord excision at the inguinal canal may be added after proper prior consent.

Hernioplasty

Definition and mechanism

- It is strengthening of posterior inguinal wall in case of indirect hernia or in any large hernia with weak abdominal wall using a supportive material.
- This allows and supports good fibroblast proliferation, which in turn strengthens the weak posterior wall of inguinal canal or abdominal wall.

Material Used

- **Synthetic:** Non-absorbable e.g. Prolene mesh, Dacron mesh, Marlex mesh, Mersilene mesh, or absorbable e.g. Vipro mesh and Ultrapro mesh.

Indications

- Direct hernia.
- Recurrent / re-recurrent hernia - incisional hernia.
- Old age.
- Hernia with weak abdominal muscle tone.
- Sliding hernia.

Principles

- Size of the mesh should be bigger than the size of the defect.
- Mesh should be fixed above to the conjoint tendon & below to the inguinal ligament using interrupted, non-absorbable sutures.
- Absolute hemostasis and prevention of infection are important.

Complications

- Infection.
- Mesh extrusion (external).
- Foreign body (FB) reaction.
- Mesh erosion into the UB, bowel or vessels may occur (*rare*).
- Mesh *inguinodynia* causing hyperesthesia & pain along the distribution of the ilio-inguinal or ilio-hypogastric nerves.

4. SLIDING HERNIA (HERNIA-EN-GLISSADE)

Definition

- The posterior wall of the *sac* is not only formed by the parietal peritoneum, but also by sigmoid colon with its mesentery on left side; cecum on right side and often with portion of the urinary bladder (Both sides) (1:2000). Rarely, small bowel sliding hernia or sacless sliding hernia can occur (1:8000). *Content* of the sac is usually small bowel or omentum.

Clinical Features

- Gender: Sliding hernia occurs exclusively in males; mainly on the left side (85%).
- Age: It is seen commonly in adults and elderly patients.
- It commonly occurs in *indirect sac*, but femoral & direct sliding hernias may occur.
- Bilaterality is extremely rare.
- Large globular swelling in the inguinal region descending into the scrotum, often irreducible or does not get reduced completely.

Treatment

- The *posterior wall of the sac should not be separated from the large bowel or bladder*. If tried, injury may result to these organs leading to fecal or urinary fistulas.
- The *partially excised sac* is pushed into the peritoneal cavity with posterior wall & repair is done usually using *prolene mesh* (Hernioplasty).
- In **LaRoque repair**, the DIR is incised above and below; reefing of the sigmoid colon mesentery is done, which forms the posterior wall of the inguinal canal so that the entire sigmoid is replaced into the peritoneal cavity.
- Right-sided sliding hernia will have the cecum and appendix in its posterior wall; the cecum should *not* be separated from posterior wall of the sac, which may otherwise → fecal fistula. The appendix should *not* be removed as it may precipitate sepsis.
- Appendices epiploicae of the sigmoid colon should *not* be removed since they may contain small colonic diverticula which may get opened to contaminate the field.
- The urinary bladder (UB) may be present on medial side of the sac, which should *not* be separated; if UB injury occurs it should be sutured in two layers with Vicryl. Inside-out purse-string suture on the opened sac is done and the sac with its posterior wall is pushed into the abdominal cavity.
- Urinary catheterization is a must and a *truss* should be completely *avoided*.

5. PANTALOON HERNIA (DOUBLE HERNIA, SADDLE HERNIA, ROMBERG HERNIA)

- Here both direct & indirect inguinal sacs are present & *straddle* the IEA.
- It clinically presents as a *direct hernia*.
- During surgery, the indirect sac may be missed, which leads to *recurrent hernia*.

6. INFANTILE HERNIA

- The processus vaginalis is closed at DIR and the hernial sac either invaginates the processus vaginalis as “inverted umbrella” or comes behind processus vaginalis.
- It is difficult to diagnose clinically, and is often diagnosed in the operating room.

Complications of Inguinal Hernia

Incarcerated Hernia

- The lumen of portion of the colon occupying a hernial sac is blocked with *feces*. Here the content of the bowel could be indented with the finger, like putty.
- In incarcerated hernia, sac and contents are densely adherent to each other (contents are fixed to the sac). It is *always irreducible; often obstructed* but may not be strangulated.

Strangulated Hernia

Pathology

- It occurs when *blood supply of the contents of hernia is seriously impaired* leading to formation of gangrene (Figure 13).
- Initially venous return is impaired → congestion of the bowel → further dilatation of the bowel, which becomes purple colored → fluid collects in the sac → eventually, arterial blood supply is impaired → bowel becomes dark, brownish black with flabby and friable wall → bacteria migrate trans-serosally & multiply in fluid of the sac → perforation occurs at the site of constriction ring → *Peritonitis occurs*.
- Common bacteria in strangulated hernia include *E. coli*, anaerobic streptococci, anaerobic bacteria and *Klebsiella*
- Strangulation commonly occurs in the *small bowel* and also in the large bowel. Occasionally, strangulated *omentocoele* also can occur *without* any intestinal obstruction.
- Strangulation can occur in inguinal, femoral, obturator, umbilical or any hernia. The IIH is more prone for strangulation than DIH due to narrow neck, adhesions and arrow SIR.

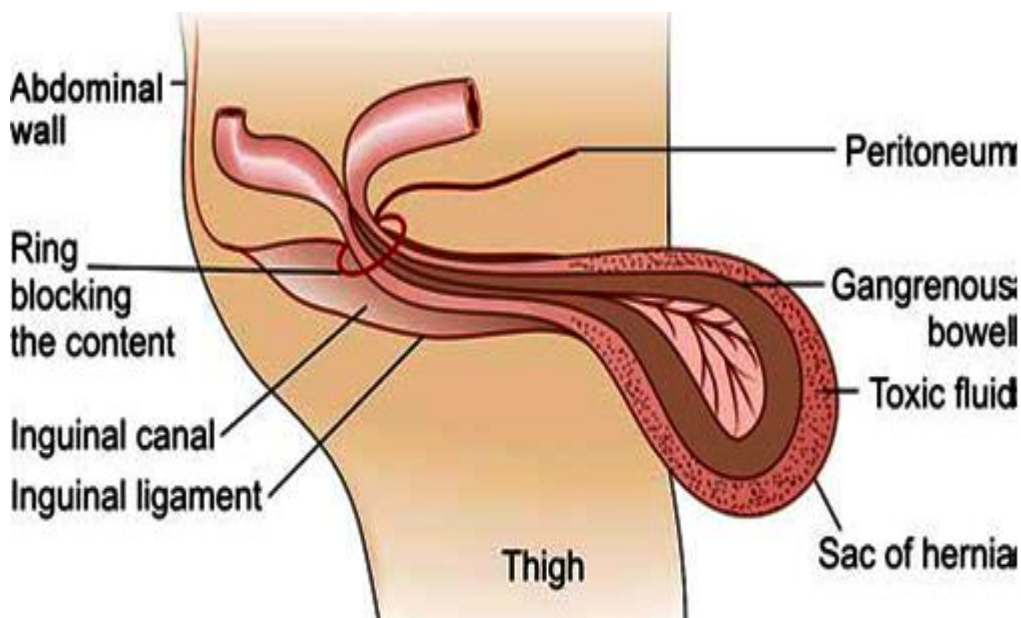


Table 13.
Strangulated
hernia with
gangrenous
bowel and
toxic fluid

- Part of circumference of the bowel when strangulated, is called **Richter's hernia** wherein the patient presents with diarrhea, toxicity mimicking *gastroenteritis*. Richter's hernia is more common with femoral and obturator hernias.

- *Maydl's hernia (Hernia-in-W)*: In this case, a loop of bowel in the form of 'W' lies in the hernial sac and the center portion of the 'W' loop is strangulated & lies within the abdominal cavity (Figure 14). Thus local tenderness over the hernia is *not* marked & the hernia gets reduced with the strangulated loop in the center of the "W". Strangulation in this case is often missed during surgery & may lead to peritonitis due to retained gangrenous loop.

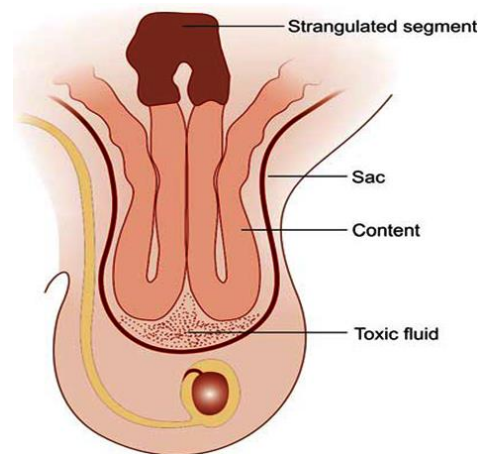


Figure 14. *Maydl's hernia (Hernia-in-W)*

Clinical Features

- *Symptoms*: Sudden severe pain, initially over a pre-existing hernia, which later becomes generalized over the abdomen. Vomiting, constipation & distension of the abdomen.
- *Local examination*: The hernia is (1) *tense*, (2) *severely tender*, (3) *irreducible* and (4) *without any expansile impulse on coughing*. *Rebound tenderness* is diagnostic.
- *General features*: Toxicity + dehydration & oliguria + electrolyte imbalance.

Investigations

- *Laboratory tests*: Serum electrolytes, serum urea and creatinine, CBC (leucocytosis)
- *Imaging*: *Plain X-ray abdomen* (erect) shows multiple air-fluid levels + US abdomen.

Treatment

- *Hospitalization*: The patient must be admitted to hospital.
- *Pre-operative medical treatment*
 - o Naso-gastric aspiration + IV fluids to correct dehydration and electrolyte imbalance.
 - o Prophylactic antibiotics.
 - o Catheterization to maintain an adequate urine output.
- *Emergency surgery*
 - o Groin incision is made with incision extending into the most prominent area of the swelling. The sac is exposed & the constriction ring and SIR are released (cut).
 - o The sac is opened carefully *without allowing the spillage of fluid* (usually spillage occurs extra-peritoneally) and the fluid is sucked with the bowel being held with fingers to prevent it from getting reduced.
 - o The viability of the bowel is checked by *color, peristalsis, pulsation, and bleeding*. If gangrenous, resection and anastomosis is done and a drain is placed.
 - o *Bassini's repair* is done by placing interrupted non-absorbable sutures. Antibiotics, IV fluids are continued. The drain is removed in 4-5 days. Once bowel movement begins, oral diet is started (in 5 days). Usually a mesh is not used in strangulated hernia repair
- *Post-operative Problems*: infection, leak with fistula, septicemia

FEMORAL HERNIA

Surgical Anatomy of the Femoral Canal

- It is the medial, most compartment of the *femoral sheath*, which extends from femoral ring above to saphenous opening below (Figure 15).
- It contains fat, lymphatics and LN of **Cloquet**.
- It is 1.25 cm long and 1.25 cm wide at the base. Below, it is closed by cribriform fascia.
- The *femoral ring* is bounded anteriorly by inguinal ligament; posteriorly by ilio-pectineal ligament of **Cooper**, pubic bone & fascia covering the pectineus muscle; medially, by the concave, sharp lacunar (**Gimbernat's**) ligament; and laterally, by a thin septum separating from femoral vein (Figure 16).

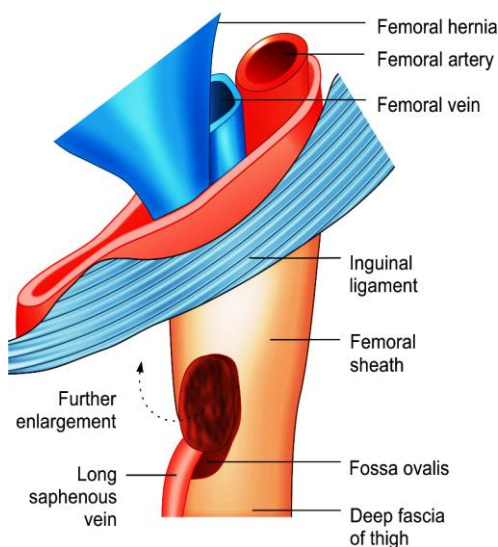


Figure 15. Anatomy of the femoral sheath

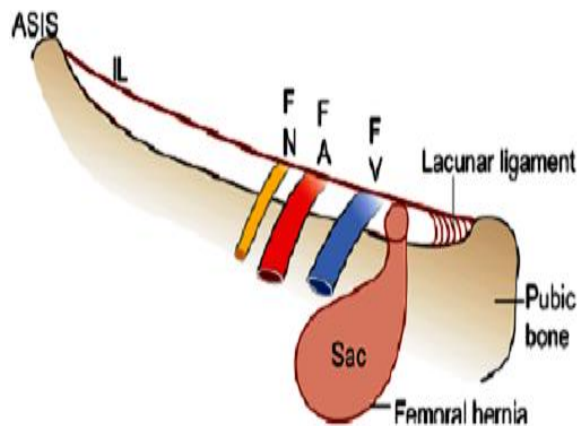


Figure 16. Anatomy of the boundaries of the femoral ring

Etiology (Predisposing Factors)

1. Wide femoral canal.
2. Multiple pregnancies.

Pathology in Femoral Hernia

- Through femoral canal, hernial sac descends down vertically up to saphenous opening and then escapes out into the loose areolar tissue to expand out like a **retort**. Because of its irregular pathway and narrow neck, it is more prone for obstruction and strangulation.
- During surgery, precaution should be taken about the femoral vein and pubic branch of obturator artery (or accessory obturator artery), which often may get injured leading to torrential hemorrhage.

Clinical Features

- It is common in females (2:1 ratio) and in multipara. It is rare before puberty.
- 20% occurs bilateral; however, more common on right side.
- Patients Present with a swelling (Figure 17) in the groin *below* and *lateral* to the pubic tubercle (inguinal hernia is *above* & *medial* to the pubic tubercle) (Figure 18).
- Swelling, impulse on coughing, reducibility, gurgling sound during reduction, dragging pain, are the usual features.



Figure 17. Right Femoral hernia presenting with a swelling in the groin

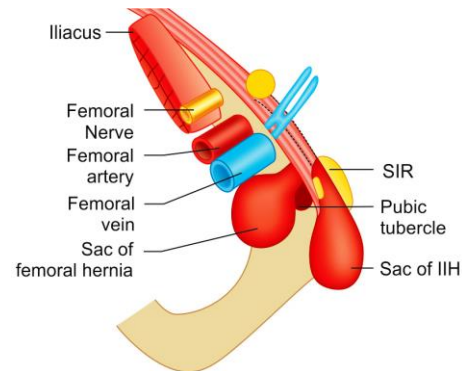


Figure 18. Femoral hernia, below and lateral to the pubic tubercle (unlike inguinal hernia)

- Obstruction or strangulation → intestinal obstruction and a painful, tender, inflamed, irreducible swelling without any impulse + features of toxicity.
- Often, a femoral hernia can be associated with inguinal hernia. Moreover, 40% of femoral hernias present as emergency hernia with obstruction/strangulation.
- **Gaur's sign:** In femoral hernia, distention of superficial epigastric and/or circumflex iliac veins occurs due to the pressure by the hernial sac.
- **Hydrocele** of femoral hernia occurs when adherent omentum secretes fluid into the sac.
- **Herniation** through a gap in the lacunar ligament (medial) is always strangulated is called **Laugier's femoral hernia**. In congenital dislocation of hip, femoral hernia occurs *behind* the femoral vessels = **Narath's femoral hernia**.
- If the sac lies under the pectineal fascia, it is called **Cloquet's hernia**.
- Often on the medial side, a portion of the urinary bladder forms the wall of the femoral hernial sac (**sliding femoral hernia**).

Treatment (Figures 19 and 20)

1. Lockwood-low Operation:

- The sac is approached *below* the inguinal ligament through a groin crease incision (or over the swelling) so that the fundus of the sac is dissected by direct vision and repair is done from below.
- *The inguinal ligament is sutured to Cooper's ligament* (standard and ideal (Cooper's ligament repair)).

2. McEvedy-high Operation

- An incision is made over the femoral canal extending *vertically* above the inguinal ligament. The sac is dissected from below, the neck from above and repair is done from above.
- It gives a very good exposure of the sac and repair is also easier. It is done in *strangulated* femoral hernia.

3. Lotheissen's Repair

- It is done through an inguinal canal approach (like inguinal hernia). The transversalis fascia is opened & the neck of the sac is identified in the femoral ring. The sac is dissected from above, the neck is ligated and repair is done. After herniotomy, *the conjoined tendon is sutured to the ilio-pectineal ligament* by interrupted sutures (2 or 3), using non-absorbable mono-filament sutures. Care should be taken to avoid injury of the femoral vein, pubic branch of obturator artery or UB.
- It is *not* as strong as Cooper's ligament repair. *Complications*, in general, include bleeding hematoma or abscess.

Differential diagnosis
❖ Inguinal hernia
❖ An enlarged Cloquet lymph node of any cause
❖ Psoas abscess—psoas spasm with flexed hip but difficulty in extension
❖ Lipoma
❖ Femoral aneurysm
❖ Distended psoas bursa (Disappears on hip flexion)
❖ <i>Saphena varix</i> —it is enlarged terminal portion of long saphenous vein. It is soft, disappears on lying down, fluid thrill, impulse on coughing and venous hum on auscultation are present. There is associated varicose veins on leg
❖ Haematoma in the region

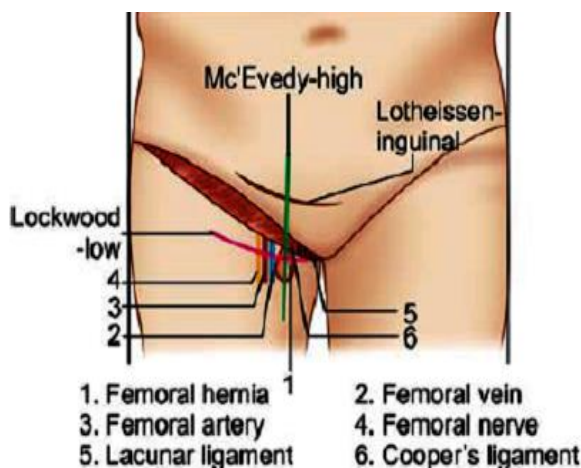


Figure 19. Different approaches for surgical repair of femoral hernia

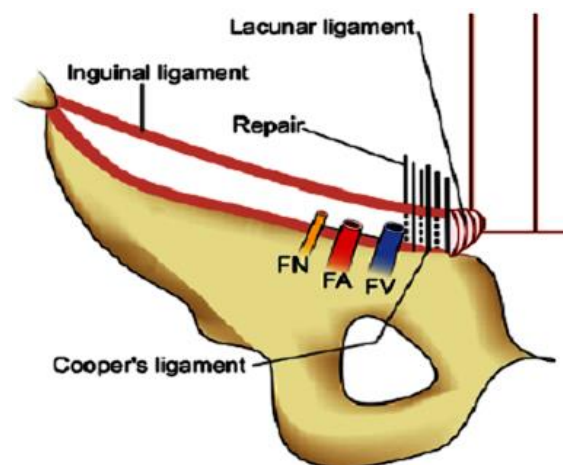


Figure 20. Femoral hernia repair (inguinal ligament to Cooper's ligament). In Lotheissen's repair, conjoint tendon is sutured to Cooper's.

4. AK Henry's Approach

- Repair of bilateral femoral hernia via a lower abdominal incision.

5. **Polypropylene mesh** can be buttressed over the femoral canal to close the defect.

6. **Laparoscopic Mesh Repair** — TEP/TAPP.

Ventral Hernias

INTRODUCTION

Definition

Any protrusion through the abdominal wall with the exception of hernia through the inguino-femoral region is defined as *ventral hernia*. Incisional hernia (80%) & primary defects in the abdominal fascia (Figure 21), which can cause umbilical hernia, epigastric hernia (Figure 22), para-umbilical hernia (PUH) or **Spigelian** hernia (Figure 23) are grouped under ventral hernia

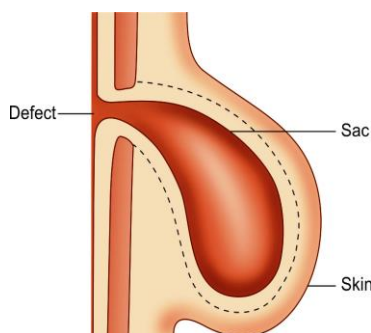


Figure 21. Fascial defect causing ventral hernia

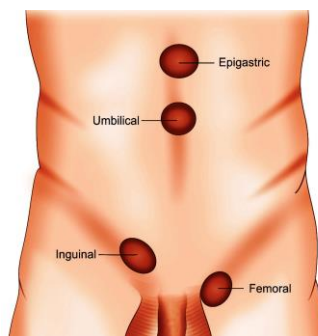


Figure 22. Epigastric AND umbilical hernias.

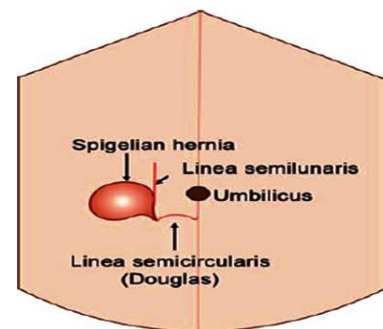


Figure 23. Classical site of Spigelian hernia

Clinical Presentation

Ventral hernia can be:

- Non-complicated (*reducible*) or *complicated* (irreducible, obstructed, strangulated).
- *Single* or *multiple* small defects (*Swiss-cheese hernia*).

Causes

- *Congenital* defect.
- *Acquired* causes e.g. Obesity, smoking, or chronic cough (straining).

Management

Pre-operative preparation

- Proper skin hygiene, respiratory care, control of obesity & antibiotics
- Relevant investigations

Surgical repair

- Open mesh repair: inlay; onlay; sublay, underlay
- Laparoscopic mesh placement: underlay (directly in front of contents) using dual mesh or four-layered mesh.

INCISIONAL HERNIA

Definition

- It is herniation through a weak abdominal scar (scar of previous surgery other than hernia repair, otherwise, it is called *recurrent hernia*).

Incidence

- It is common in old age and obese individuals.
- It occurs in 10% of abdominal surgeries; 70% in the first 5 ys and 30% within 5-10 ys.

Predisposing Factors

- Abdominal scars may injure nerves of abdominal muscles e.g. scar of major surgeries (biliary, pancreatic) and scar of emergency surgeries (peritonitis, acute abdomen).
- Faulty technique of closure.
- ↑ intra-abdominal pressure (straining, cough, BPH, straining, stricture urethra, ascites).
- Poor nutritional status of the patient, anemia and hypo-proteinemia
- Debilitating disease (e.g. uremia, TB, jaundice or .malignancy)
- Immunosuppression.
- Smoking in postoperative period.

Clinical Features

History-Taking

- Details of previous surgery and duration of the incisional hernia itself should be recorded.
- History of wound infection, wound dehiscence, whether surgery done was an emergency or elective, and tension sutures placed or not.
- History of pain, irreducibility and details of precipitating factors should be recorded.
- Other precipitating factors are similar to inguinal hernia like smoking, urinary / respiratory / abdominal symptoms.

Clinical Examination

- Swelling in the scar region. It may be *small or large; huge or massive (diffuse)* (Figure 24), more common in the lower abdomen.
- Impulse on coughing.
- Gurgling sound ± visible bowel peristalsis
- Eventually features of irreducibility, obstruction, strangulation (complications)
- Scar, its site and extent, whether healed primarily or secondarily, skin over the scar and swelling is noted.
- Details of the swelling with expansile impulse on coughing and examination (*supine/standing*) are done.
- *The gap cannot be assessed in irreducible hernia.*



Figure 24. Huge incisional hernia. Note scar of previous operation

Investigations

The precipitating factors must be looked for:

- Imaging: Chest X-ray - US abdomen.
- Tests relevant for causes.

Pre-operative Preparation

- Reduction of weight and control of obesity.
- Correction of nutritional deficiencies and anemia.
- Treatment of diabetes mellitus (DM), hypertension, cardiac and respiratory problems or other precipitating causes.
- Massive incisional hernia after reduction might cause IVC compression, paralytic ileus and diaphragmatic elevation with respiratory embarrassment (*abdominal compartment syndrome*). It is prevented by prior increasing the capacity of peritoneal cavity by creating pneumo-peritoneum using CO₂ so as to increase the peritoneal pressure by 12-15 cm of H₂O, daily for 3-6 weeks. Later, definitive surgery is done.

Treatment Strategy

- *Mesh repair of the incisional hernia* defect is always better and ideal with less recurrence.
- Adequate-sized *mesh* is placed either outer to the peritoneum (*sublay*), or outer to the musculo-aponeurotic abdominal layer (*onlay/overlay*), or occasionally combined sublay and onlay mesh placement, both deeper and outer to the musculo-aponeurotic layer.
- **Rive's Stoppa's mesh placement** for incisional hernia is placing the mesh between the posterior rectus sheath & rectus muscle (*retro-rectus mesh placement* is easier for dissection and mesh placement).
- Commonly Polypropylene mesh is used. Other materials used are Dacron, Polytetrafluoroethylene (PTFE) mesh, Polyglycolic mesh (Vicryl mesh) or combined Polypropylene and Polyglycolic acid mesh (Vipro mesh). Prevention of infection/hemostasis and drainage are important. A *drain* (suction drain) must be placed after surgery.
- **Laparoscopic mesh repair** is done for incisional hernia by placing a mesh *under* the defect *laparoscopically* in the *intra-peritoneal* plane. The problem of this underlay placement is chances of adhesion and GI fistula formation, but still it is found to be safer. Laparoscopic *pre-peritoneal* mesh-placement is also done for smaller defects. Currently, *dual mesh* (PTFE) or 4-layered mesh is available. Here, the mesh is placed *under the peritoneum* deep to the defect after reducing the contents. The mesh is fixed with sutures and tacks. In four-layered mesh, the deepest 1st layer is made of absorbable cellulose, which allows new peritoneum to creep underneath. The 2nd layer is PDS/PTFE mesh, the 3rd layer is Polypropylene mesh and the last 4th layer is again PDS/PTFE mesh. It is ideal but *costly*.
- **Cattell's operation:** When the defect is <3 cm and if the patient is having adequate abdominal muscle tone, *layer-by-layer anatomical repair* is done using monofilament non-absorbable suture material like *Polypropylene / Polyethylene*, ideally with interrupted sutures. The sac should be dissected, ligated & excised prior to repair. The peritoneum and posterior rectus sheath are opposed as 1st layer & the anterior rectus sheath as 2nd layer.

- **Double breasting of the rectus sheath** using interrupted non-absorbable sutures using mono-filament suture. It is overlapping the rectus sheath in 2 layers with 2 rows of sutures.
- **Keel operation** is done in large defects. The scar is excised and dissected beyond the margin of the defect. The sac is *never* opened unless there is obstruction of the content. The sac is inverted using continuous/interrupted inverting non-absorbable sutures, layer-by-layer until the defect margins are opposed together, which is then again sutured with interrupted sutures. (N.B. *Keel is inverted beam of the ship*).
- **Nuttall's operation** is done for lower midline incisional hernia. Attachments of the recti muscles are detached from the pubic bones and are crossed over to fix to opposite pubic bones so as to create a firm abdominal wall support by crossed recti muscles.

Different Types of Mesh Repair for Incisional Hernia

- *Sublay (outer to the peritoneum) is the ideal method.* Large-sized mesh is placed pre-peritoneally. It needs *not* be fixed as abdominal pressure keeps it in position (according to *Pascal's law*).
- *Underlay (under the peritoneum, directly over the content):* Now it is accepted, but there are chances of adhesions/fistula formation. It is used in laparoscopic repair. Dual mesh/four-layered mesh is used.
- *Overlay mesh (outer to the musculo-aponeurotic layer).* Here, the mesh is placed under the SC tissue. It carries a high recurrence rate (RR) (30%). Thus, it is *not recommended*.
- *Combined inlay and overlay* with 2 layers of mesh.
- *Rive's Stoppa's method* of placing the mesh between the posterior rectus sheath & the rectus muscle widely. Dissection is done at least 10 cm beyond the defect under the rectus muscle in front of the posterior rectus sheath; the mesh should cover 5 cm beyond the defect all around. Recurrence rate (RR) with this method is <10%.
- *Components separation technique* is a better method in large defects. Here, the skin and SC fat are dissected of the anterior rectus sheath → EOA is incised 2 cm lateral to the rectus abdominis → the EOA is separated from the internal oblique until posterior axillary line → additional relaxing incisions are made over the rectus sheath, internal oblique, transverse abdominis → it is repeated on opposite side → 20 cm mobilization is achieved → also reinforced with a large mesh. Component separation technique causes often lateral bulges on both sides. If it is done without mesh support and RR is high. Its *advantage* is that a defect up to 20 cm can be easily brought together. The technique is also called as *Autologous repair by vascularized innervated muscle flaps* (**Ramirez, 1990**).
- *Inlay (bridging at myo-aponeurotic level outer to peritoneum):* Adequate-sized mesh is placed in the defect level bridging the gap to cover the pre-peritoneum. It is better by principle but still shows a high RR.

Surgeries Recommended for Repair of Incisional Hernia

- Mesh repair — commonly done and ideal
- Component separation technique
- Layer-by-layer closure — Cattell's operation
- Double-breasting of the rectus sheath

Post-operative Care

- Analgesics and antibiotics.
- Nasogastric aspiration and prevention of paralytic ileus + fluid management
- Abdominal binder for support of the abdominal wall during recovery period
- Control of obesity & other precipitating factors.
- Stopping smoking and treating other associated causes.
- Early ambulation.
- The drain should be kept until drainage becomes minimal.

Complications of Incisional Hernia Surgery

- Wound infection, sinus or seroma formation
- Paralytic ileus
- Abdominal compartment syndrome in large hernias
- Entero-cutaneous fistula
- Infection of the mesh
- Recurrence

Additional Problems in Large Incisional Hernia

- While reducing the bulky contents like bowel and omentum, inadequate intra-abdominal capacity results in increased intra-abdominal pressure causing inferior vena cava (IVC) compression, mesenteric edema followed by stasis of splanchnic bed, paralytic ileus, diaphragmatic elevation and respiratory distress (*abdominal compartment syndrome*), urinary and bowel disturbances. Abdominal capacity can be raised by creating regular pneumo-peritoneum over the period of 3-6 weeks.
- Lordosis and back pain may be the presenting feature.
- The sac and contents may get adherent to the thin skin over the summit of the hernia leading to skin ulceration and occasionally, fistula formation.
- It may often require resection of the adherent bowel segment.
- Large mesh-placement is required.

Important Notes

- *It is now universally accepted that prosthetic repair is the gold standard for all incisional hernia.*
- Non-prosthetic repair has got only historical value as the recurrence rate is very high (as high as 55%).

UMBILICAL HERNIA

Anatomy of Umbilical Region

- The *umbilical ring* is a complex structure, which is related to the linea alba, falciform ligament, obliterated urachus (median umbilical ligament) and umbilical fascia (*Richet's fascia*).
- It is the meeting point of 4 folds of embryonic plate and three systems; GIT (vitello-intestinal), urinary (urachus) and vascular (umbilical vessels).
- It is located at the level of L4 and L5 vertebral discs; at a lower position in infants.
- It is a water-shed area for venous and lymphatic drainage; above the umbilicus, it drains to axillary vein or lymphatics; and below to the inguinal area.
- Umbilical skin is supplied by T10 spinal cord.
- Umbilical hernia develops due to either absence of umbilical fascia or incomplete closure of umbilical defect. The weakest part in the umbilical cicatrix is the upper part where hernia begins.

Etiology

- Umbilical hernia is herniation through a weak umbilical scar (cicatrix).
- It can be *congenital* in newborns and infants (common in males, 2:1) or *acquired* in adults (common in females).
- Congenital umbilical hernia is *common* in Africa or in people of African origin (8 times), while the acquired type is similar to any ventral hernia.
- It is common in infants and children, occurs commonly due to neonatal sepsis.
- It is seen in 20% of newborn infants.
- Umbilical hernia is common in *Down's syndrome*, *Beckwith-Weidman syndrome*.

Clinical Features

- It presents with a swelling in the umbilical region (Figure 25) within the first few months after birth, the size ↑ during crying. It is hemi-spherical in shape.
- The defect can be felt with the finger during crying.
- It can undergo irreducibility and obstruction, which presents with pain, distension, vomiting.

Treatment

- **Initially, conservative.** In 93-95% of cases, it disappears spontaneously in few months after birth (*masterly inactivity*). It can be hastened by adhesive strapping across the abdomen.



Figure 25. Umbilical hernia

Indications for Surgery

1. If it persists; even after the age of two years.
2. If the defect is > 2 cm in size.
3. Acquired/adult umbilical hernia.

Surgical Procedures

1. Primary closure of the defect

- An infra-umbilical incision (Figure 26) is made encircling its lower half (*smiling incision*).
- The sac is dissected circumferentially and is released off from the umbilicus and SC tissue. It is opened; contents are reduced; excess part is excised up to the umbilical ring. The defect is closed with interrupted non-absorbable Prolene sutures.

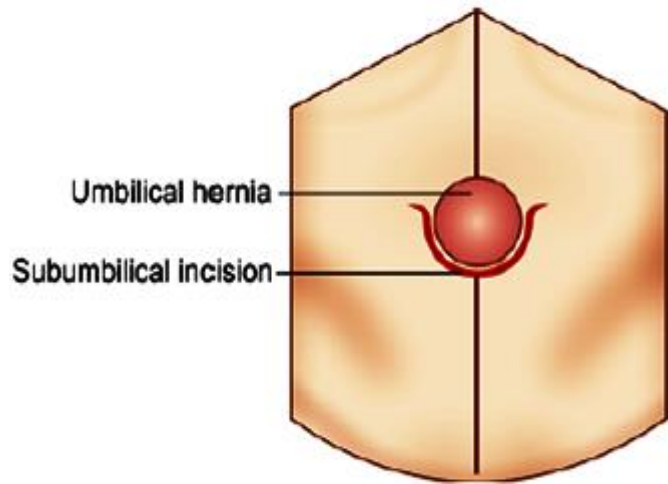


Figure 26. Infra (sub)-umbilical incision for surgical treatment of umbilical hernia

2. Mayo's operation

- The horizontally-opened rectus sheath is approximated as double-breasting with the upper flap overlapping the lower flap in front. It is rarely done today.

3. Sub-lay mesh repair

- In a large umbilical hernia (defect >3 cm) with a degenerated skin on its surface, it is often difficult to retain the umbilicus. When it is attempted to save the umbilicus, the infra-umbilical incision should extend laterally about 2 cm on each side at 3 and 9 O'clock positions. The sac is dissected and is excised after excision of redundant part.
- Currently, it is standard to use polypropylene mesh as sub-lay or in the retro-rectus position and then the rectus sheath is closed.

4. Umbilectomy

- Unhealthy thin skin over a large umbilical hernia is a real problem. It is better to do umbilectomy in these patients (excision of umbilical cicatrix).
- It is done *only* in *adults* with large umbilical hernia with thinning of the umbilical skin. Prior consent and repair and creation of the umbilicus (*umbilicoplasty*) is needed.

5. Open dual PTFE and Polypropylene mesh placement

- Umbilical hernia is dissected similarly through a sub-umbilical incision. Redundant sac is excised and the peritoneum is not closed. A special composite mesh containing wider PTFE on the inner side with little smaller Polypropylene mesh on the outer aspect is used. It has also got two additional straps of the mesh attached to the outer part of the main composite mesh which are used to hold and later to fix into the defect anteriorly.
- This technique yields excellent results with a low recurrence rate.

6. Laparoscopic umbilical hernia repair

- It is similar to any ventral hernia repair through laparoscopy, which is done under general anesthesia. It is useful only in *large* umbilical hernia.

PARA-UMBILICAL HERNIA (PUH)

Etiology / Pathology

- It is a protrusion or herniation through the linea alba, just *above or below* the umbilicus.
- It enlarges ovoidly, often attaining a large size and sags downwards.
- The neck of the sac is relatively narrow.
- Contents are usually omentum, small bowel and sometimes, large bowel.
- It has a tendency to undergo adhesion formation, irreducibility & obstruction.

Clinical Features

- **Age:**
It occurs commonly in adults
- **Gender:**
Common in females > males (5:1).
- It presents as a swelling (Figure 27), which has a smooth surface, distinct edges, soft consistency, resonant on percussion (bowel) with dragging pain and impulse on cough. Large hernias can present with intestinal colic due to subacute intestinal obstruction. Eventually, obstruction and strangulation can occur.



Figure 27. Para-umbilical hernia (PUH) in an adult gentleman.

Treatment

Treatment is Always Surgical

- Dissection of the hernial sac and placement of mesh in the retro-rectus plane and under the umbilicus is the ideal treatment.
- Often, *umbilectomy* is required and *mesh placement is useful* when the defect is large (> 4 cm).
- If there is strangulation, resection of bowel segment and anastomosis is done followed by repair of the hernia.
- ***Mayo's operation***
 - Via a transverse elliptical incision, the sac is identified and dissected.
 - Herniotomy is done.
 - *Double-breasting of the defect* in the rectus is done by interrupted non-absorbable sutures.
- Additional lipectomy (*panniculectomy*) may be done in case of obese patients with pendulous abdomen.
- Post-operative weight reduction.
- The use of abdominal binder is required for these patients.

FATTY HERNIA OF THE LINEA ALBA / EPIGASTRIC HERNIA

Pathology

- It occurs usually through a defect in the decussation of the fibers of the linea alba anywhere between the xiphoid process and the umbilicus.
- Extra-peritoneal fat protrudes through the defect as *fatty hernia of the linea alba* presenting like a swelling in the upper midline with an impulse on cough (Figure 28)
- *It is a sacless hernia*. Later, protrusion enlarges and *drags a pouch of peritoneum*, presenting as a *true epigastric hernia*.
- If the defect is <1.5 cm, the lateral margin of the defect is formed by only anterior and posterior lamina of the rectus sheath; if the defect is >1.5 cm, then lateral margin is formed also by the rectus muscle.
- *Content* of true epigastric hernia is omentum, but sometimes it may be small bowel.

Clinical Features

- Common in muscular men / manual laborers.
- 20% of epigastric hernias are multiple (*Swiss-cheese-like*).
- It is often *asymptomatic*.
- Swelling in the epigastric region, which is tender.
- Pain in epigastric region; often associated with peptic ulcer (may be the only cause of pain), which should be ruled out by gastroscopy.
- Impulse on cough and defect in the epigastric region are also positive.
- Irreducibility, obstruction, strangulation as seen in any other hernia.

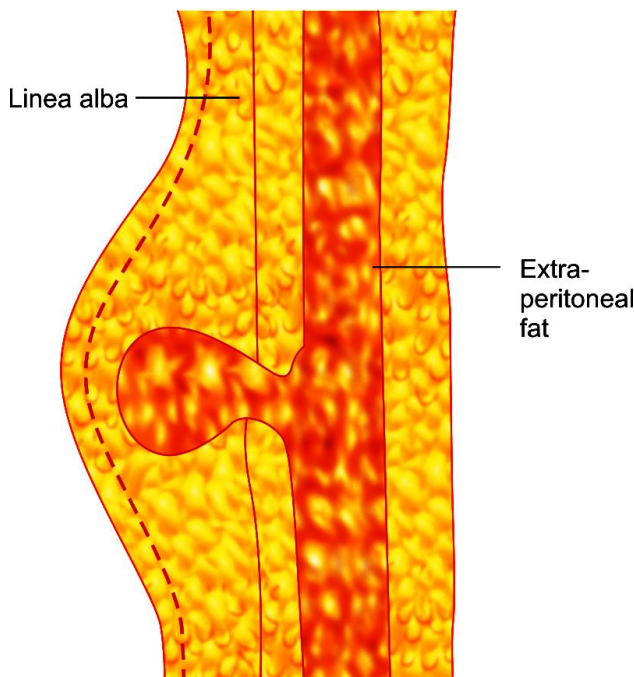


Figure 28. Fatty hernia of the linea alba (FHLA). Just protrusion of the extra-peritoneal pads of fat occur i.e. NO sac.

Treatment

- Through a vertical incision, the sac is dealt with and the defect is closed as usual.
- A large defect is supported with a pre-peritoneal mesh.
- Complete reconstruction of the linea alba is needed from the xiphi-sternum to the umbilicus in *Swiss-cheese* type using different methods such as interrupted 1^{ry} closure using
- Polypropylene sutures or the *modified shoelace technique* (after removing strip of the medial margins of the linea alba; double-breasting of the linea alba is under-taken).

SPIGELIAN HERNIA

Definition and Site

- It is a type of *inter-parietal hernia* occurring at the level of the *arcuate line* through the Spigelian point.
- The hernial sac lies either deep to the internal oblique muscle or between the external and internal oblique muscles.

Surgical Anatomy and Pathology

- It is a lateral ventral hernia through ***Spigelian fascia*** at any point along its line.
- *Semilunar line of Spigel* is a line from pubic tubercle to tip of 9th costal cartilage. It marks the lateral margin of the rectus sheath.
- Semicircular arcuate line (fold) of ***Douglas*** is the lower end of posterior lamina of rectus sheath below the umbilicus & above the pubis.
- *Spigelian fascia* is the area between the lateral border of the rectus muscle and the external and internal oblique and transversus abdominis muscle.
- Spigelian hernia can occur above (10%) or below (90%) the umbilicus. Below the umbilicus, it occurs at the junction of linea semilunaris and the linea semicircularis (wider and weaker point).
- In Spigelian hernia, the defect is formed by the internal oblique & transversus abdominis muscle. The external oblique is *outer* to the hernial sac.

Clinical Features

- Age and gender: It is common in females >50 y.
- It presents as a soft, reducible mass *lateral* to the rectus muscle and *below* the umbilicus, with impulse on cough.
- Strangulation is common.
- Presence of precipitating factors (obesity, chronic cough, old age, multiple pregnancies).

Differential Diagnosis

- It should be differentiated from:
 1. Abdominal wall lipoma
 2. Hematoma
 3. Soft tissue sarcoma.

Treatment

- Through a lengthy transverse incision, herniotomy & later, closure of the defect layer-by-layer using non-absorbable interrupted sutures. But, *ideally* a *mesh* is required to cover the defect properly.
- Laparoscopic dual mesh placement is also useful.

COMPLICATIONS OF HERNIA SURGERY

- Twenty years ago when tissue repair was popular, *recurrence* was the most worrisome complication. Nowadays, since recurrence rate has been reduced with the standard usage of mesh, *mesh-related complications* have become more common than recurrence.
- Inguinodynia (pain in the inguinal region), mesh extrusion, mesh infection, mesh erosion are now the worrisome problems.
- Complications of hernia surgery can be classified as *early*, *delayed*, *late*, or *life-threatening*.
- According to technique, complications can be classified into those related to *open surgery* or *laparoscopic* (TEP/TAPP).

Complications of Open Hernia Surgery	Complications of TEP/TAPP
- Infection - Seroma / hematoma formation - Groin pain – osteitis pubis - Inguinodynia (neural) - Ischemic orchitis (thrombosis of pampiniform venous plexus (0.5%). - Injury of vas deferens – injury of viscera - Hydrocele formation - Recurrence - Dysejaculation: Painful, burning sensation just before, during or after ejaculation due to cremaster dysfunction or vas stricture (0.25%).	- SC emphysema - Pneumothorax - Hyper-carbia - Vascular - Neural - Visceral - Infection - Ileus - Conversion - Recurrence

Inguino-Scrotal Swellings Other than Hernias

CAUSES

- Inguino-scrotal swellings almost necessarily arise in the constituents of the spermatic cord or in a persistent funicular process.
- The following must be considered:
 1. Hernia (reducible, irreducible, strangulated) (**Refer back**)
 2. Hydrocele (congenital hydrocele, infantile hydrocele, hydrocele of the cord, hydrocele of the hernial sac).
 3. Varicocele.
 4. Other spermatic cord lesions (causing thickening of the cord):
 - a) Traumatic (hematocele of cord).
 - b) Inflammatory.
 - c) Malignant thickening of the cord.
 5. Testicular causes.

HYDROCELE

Types of Hydrocele that Present with Inguino-scrotal Swelling

Congenital Hydrocele

- It is a persistence of the processus funicularis throughout its length, and differs from a congenital hernia in that its communication with the abdomen is shut off, or so nearly shut off that bowel or omentum *cannot* enter it (Figure 1).
- The swelling fills the tunica vaginalis and obscures the testis, and continues as a tubular prolongation up the inguinal canal. It is elastic or fluctuant, and trans-illuminates easily.
- It is differentiated from a hernia by its translucency, and absence of both impulse on cough and reducibility.
- Though irreducible (digital pressure on the hydrocele does not usually empty it), many congenital hydroceles have a small communication with the abdomen indicated by its reduction in size during the hours of sleep.
- The appearance of bilateral congenital hydrocele in a baby is often an index of peritoneal mischief, such as tuberculous peritonitis.

Infantile Hydrocele

- It occurs when a completely patent processus funicularis remains unobserved in infancy and becomes distended during adult life. The tunica and processus vaginalis are distended to the inguinal ring, but there is no connection with the peritoneal cavity
- It has the same physical characters mentioned above, but it does *not* vary in size.

Encysted Hydrocele of the Cord

- It occurs if the processus funicularis is shut off from both, the tunica vaginalis and peritoneum, leaving a patent portion in between, which becomes distended with fluid (Figure 1).
- It may give an *inguinal, scrotal or inguino-scrotal* swelling depending on the patent part of the funicular process.
- There is a smooth oval swelling near the spermatic cord, which is liable to be mistaken for an inguinal hernia. The swelling is characteristically cystic and translucent.
- Traction test is positive. The swelling moves downwards and becomes less mobile if the testis is pulled gently downwards



Figure 1. Hydrocele

Hydrocele of the Hernial Sac

- There is collection of fluid in the hernial sac, the opening of which at the internal ring being closed by a plug of omentum.
- There is a history of hernia, and the testis is felt separate.

Pathogenesis

A hydrocele can be produced in four ways:

1. By connection with a hernia of the peritoneal cavity in the congenital variety, which presents as hydrocele of the cord
2. By excessive production of fluid within the sac, e.g. Secondary hydrocele
3. Through defective absorption of fluid
4. By interference with lymphatic drainage of scrotal structures (elephantiasis)

Complications

Among complications of hydrocele are the following:

1. *Rupture* usually occurs as a result of trauma, but may be spontaneous. On rare occasions cure results after the fluid has been absorbed.
2. *Transformation into a hemocele* occurs if there is spontaneous bleeding into the sac or as a result of trauma. Acute hemorrhage into the tunica vaginalis sometimes results from testicular trauma and it may be difficult without exploration to decide whether the testis has been ruptured. If the hemocele is not drained, a clotted hemocele usually results.

3. *The sac may calcify.* Clotted hydrocele may result from a slow spontaneous ooze of blood into the tunica vaginalis. It is usually painless and by the time the patient seeks help, it may be difficult to be sure that the swelling is not due to a testicular tumor. Indeed, a tumor may present as a hematocele.
4. *Infection*, which may lead to “pyocele”.
5. *Severe infection* can, occasionally, be introduced by aspiration. Simple aspiration, however, often may be used as a temporary measure in those cases where surgery is contraindicated or must be postponed.
6. *Post-herniorrhaphy hydrocele* is a relatively rare complication of inguinal hernia repair. It may be due to interruption to the lymphatics draining the scrotal contents.
7. Atrophy of testis in long standing cases.

Diagnosis

A primary hydrocele is described as having the following characteristics:

- *Transillumination*: positive
- *Fluctuation*: positive
- *Impulse on coughing*: negative (positive in congenital hydrocele)
- *Reducibility*: absent
- *Testis cannot be palpated separately*, except in funicular hydrocele, and encysted hydrocele.

Treatment

- Most hydroceles appearing in the first year of life seldom require treatment as they resolve without treatment.
- Hydroceles that persist after the first year or occur later in life require treatment only in selected cases, such as patients who are symptomatic with pain or a pressure sensation, or when the scrotal skin integrity is compromised from chronic irritation; the treatment of choice is *surgery* and the operation is conducted via an open access technique aiming to excise the hydrocele sac. Anesthesia is required for the operation and general anesthesia is of choice in children, while spinal anesthesia is usually sufficient in adults. Local infiltration anesthesia is not satisfactory because it cannot abolish abdominal pain due to traction on the spermatic cord. In long-standing cases, hydrocele fluid may be opalescent with cholesterol and may contain crystals of tyrosine.
- After *aspiration* of a primary hydrocele, fluid re-accumulates over the following months and periodic aspiration or operation is needed. For younger patients, operation is usually preferred.
- *Sclerotherapy* is an alternative; after aspiration, 6% aqueous phenol (10-20 ml) together with 1% Lidocaine for analgesia can be injected, and this often inhibits reaccumulation. These alternative treatments are generally regarded as *unsatisfactory* treatment because of the high incidence of recurrences and the frequent necessity for repetition of the procedure.

VARICOCELE

Surgical Anatomy (Veins of the Testis)

1. *Pampiniform Plexus*: Intercommunicating veins accompanying the testicular artery in the spermatic cord. They form 2-3 testicular veins in the inguinal canal and then one testicular vein at the level of the deep inguinal ring (DIR).
2. *Cremasteric Veins*: Anastomose freely with testicular veins to provide an alternative venous return. They leave the cord in the inguinal canal, piercing the posterior wall to end in the inferior epigastric vein.
3. *Veins of the Vas*: a small vein that passes the DIR, then passes medially into the pelvis to end in the inferior vesical vein and finally into the internal iliac vein.

Definition of Varicocele

Varicocele is pathological alteration in the venous drainage of the testis; defined as a state of varicosity (dilatation, elongation and tortuosity) of the testicular veins (Figure 2). Varicocele may be *primary* or *secondary*.

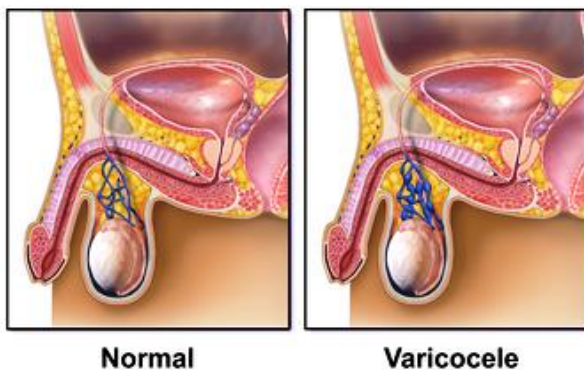


Figure 2: Varicocele: engorgement of the pampiniform plexus

Pathophysiology

- Often the greatest concern with respect to varicocele is its effect on male fertility. The relationship between varicocele and infertility is unclear. Some men with the condition are fertile, some have sperm that are normal in shape and move normally but are compromised in function, and some have sperm with abnormal shapes or that do not move well. Theories as to how varicocele affects sperm function include damage via excess heat caused by the blood pooling and oxidative stress on sperms.
- Tobacco smoking and mutations in the gene expressing glutathione S-transferase both put men at risk for infertility; these factors may also exacerbate the risk that varicocele will affect fertility.

Signs and Symptoms

- Varicocele occurs in about 15% to 20% of all men. Its incidence increases with age.
- It is an inguino-scrotal swelling (chiefly scrotal).

- It is nearly always on the left side, the patient being usually a young adult with dragging pain or heaviness in the scrotum, particularly after prolonged standing (Figure 2).
- The distended veins of the pampiniform plexus resemble a “bag of worms”. With a lax scrotum, the blue color of veins can be seen.
- A varicocele is said to have an impulse on coughing, but the sensation is that of a fluid thrill or a small tap on the examining finger rather than an impulse.
- Squeezing empties the swelling, which refills immediately. Recumbency with the scrotum raised also causes it to disappear.



Figure 2. Left varicocele

Clinical Classification (*Dubin*)

Group	Description
Group 1 (large)	- Easily visible at a distance. - Palpation reveals a bulk of dilated veins >2 cm in diameter on Valsalva's maneuver.
Group 2 (moderate)	- o Less easily seen. - o Palpation reveals a bulk of dilated veins 1-2 cm in diameter.
Group 3 (small)	- Cannot be seen by inspection. - Bulk of veins < 1 cm by palpation (but +ve reflux with Valsalva's).

Diagnosis

Following discovery of the sign of swelling comprising a mass, varicocele can be confirmed with scrotal *ultrasound*, which will show dilation of the vessels of the pampiniform plexus to be greater than 2 mm.

Complications of Varicocele

1. Atrophy of the testis due to chronic congestion (pressure and thermal effect).
2. Oligospermia (sterility, if bilateral) due to the same effects.
3. 2^{ry} hydrocele, due to chronic congestion of the testis.
4. Repeated attacks of thrombosis due to blood stagnation in the veins.
5. Redundancy of the scrotum may interfere with the patient's normal activities, and also makes the testis more liable to torsion and trauma, and may cause neurosis.

Differences between 1^{ry} & 2^{ry} Varicocele

Criteria	Primary Varicocele	Secondary Varicocele
<i>Incidence</i>	More common	Less common
<i>Etiology</i>	- Genetic factor (familial tendency). - Wall weakness - Venous hypertension (congestion).	- Abdominal mass or retroperitoneal fibrosis. - Hypernephroma invading the renal vein, or causing compression from outside.
<i>Age</i>	Young adults.	Older age (40-50 years).
<i>Side</i>	Left side (96%), right side (2%), bilateral (2%)	According to cause and site of obstruction.
<i>Onset</i>	Gradual.	Rapid and sudden.
<i>Clinical Picture</i>	1. Dragging pain. 2. Impulse and thrill on cough. 3. Bowing test +ve. 4. Softer and smaller testicle. 5. No evidence of 1^{ry} cause.	1. Painless. 2. No impulse and no thrill. 3. Bowing test -ve. 4. Larger and firmer testicle. 5. Evidence of 1^{ry} cause.

Treatment

- The two most common *surgical approaches* are retro-peritoneal (abdominal using laparoscopic surgery), infra-inguinal/sub-inguinal (below the groin), and inguinal (groin using percutaneous embolization).
- Possible *complications* of this procedure include hematoma (bleeding into tissues), hydrocele (accumulation of fluid around the affected testicle), infection, or injury to the scrotal tissue or structures. In addition, injury to the artery that supplies the testicle may occur, resulting in a loss of a testicle.

Prognosis

- Whether having varicocele surgery or embolization improves male fertility is controversial, as good clinical data is lacking. There is tentative evidence that *varicocelectomy* may improve fertility in those with obvious findings and abnormal sperm.
- Evidence for sclerotherapy is unclear as of 2015.

Important Clinical Notes

Why does 1st varicocele occur more on the left side than the right ??

1. Higher incidence of left testicular vein abnormal valves (absent, malformed or incompetent).
2. The left testicular vein ends in the renal vein at a right angle, which impedes the flow.
3. Blockage of the valve at the left testicular vein by the SMA or left colon.
4. Earlier descent of left testis → lower level in the scrotum → longer & more dependent veins.
5. *Nutcracker mechanism*: compression of left renal vein between the abdominal aorta and superior mesenteric artery (SMA).
6. Compression of left renal vein by the ligament of Trietz or arched testicular artery.
7. The left adrenal vein enters the left renal vein opposite to the stoma of the left testicular vein causing turbulence of blood and impedance of flow.

Why is spermatogenesis impaired in cases of varicocele ??

1. Disturbed scrotal *thermoregulatory system* → impaired testicular function due to ↑ intra-scrotal temperature.
2. *Mechanical pressure* of the dilated pampiniform plexus may → testicular dysfunction.
3. Stasis of blood → *hypoxia* and tissue destruction especially of the germinal epithelium of the testis.
4. Impaired *epididymal environment* which may be produced by varicocele → disturbed motility of the sperms.
5. Retrograde passage of *toxic materials* (mainly cortisol) from the left renal or left adrenal vein to the pampiniform plexus.
6. *Hormonal (chemical) factors*:
Alteration in testicular function of infertile men with varicocele > fertile men with varicocele:
 - *Catecholamines* (found more in the left testicular vein and cause spermatogenic dysfunction).
 - *Prostaglandins* (PGDs E & F) (higher levels in the left internal spermatic vein → inhibited spermatogenesis due to reduced blood flow through the testis and acceleration of contractility of the accessory sex organs, which → reduced maturity of sperms).
 - *Serotonin* → dys-spermatogenesis due to its vaso-constrictive effect on testicular blood supply.

How can you diagnosis subclinical cases ?

1. *Doppler echography*: With the patient recumbent, the probe is placed over the spermatic cord. With Valsalva's maneuver, venous pressure increases → reversal of blood flow in the testis (i.e. incompetent valves).
2. *Scrotal thermography*: Scrotal temperature reflects that of underlying structures. With venous stagnation, temperature is elevated.
3. *Retrograde spermatic venography (via a catheter in the left renal vein)*: It shows *reflux*. It is a definitive test but invasive & therefore not used for screening and has been replaced by Doppler and thermography.

OTHER SPERMATIC CORD LESIONS

Trauma (Hematocele of the Cord)

- It presents as an ill-defined opaque swelling which is tender and irreducible.
- It occurs soon after trauma such as a kick in the groin.

Inflammation

- Inflammatory thickening of the cord may be an extension of inflammation of the epididymis and testis.
- *Tuberculous funiculitis*
 - o There is generalized fullness of the cord.
 - o More typically the “beaded” vas may be felt.
- *Filarial funiculitis*
 - o It is usually acute with high fever swelling and redness of the skin.
 - o The cord becomes thickened, tender and matted.
 - o There is always a well-pronounced hydrocele.
 - o The prostate and seminal vesicles are normal.
 - o The patient usually comes from an endemic area.
 - o Blood shows leukocytosis and eosinophilia and may show microfilaria.
 - o Sometimes a residual mass is felt in the cord after the acute attack resolves.
 - o It resembles a *bilharzial mass*, but differentiation can be made by the following:

Filarial Mass	Bilharzial Mass
- Cord is matted - Mass diffusely affect the cord. - Normal prostate. - Patient comes from an endemic area. - No bilharzial ova in urine.	- Cord not matted. - Mass in the lower 2 inches of the cord. - Enlarged, firm, irregular prostate. - History of bilharziasis. - Ova in urine.

Lymph Varix (Lymphangiectasis, Diffuse Hydrocele of the Cord)

- There is dilatation and tortuosity of the lymphatic vessels of the cord, caused by obstruction due to *filaria*.
- It gives an impulse (thrill-like) on coughing and reduces spontaneously on lying down.
- It may feel soft, cystic or doughy.
- It is diagnosed by the history of periodic attacks of fever with development of swelling and tenderness of the cord, eosinophilia and living microfilaria in blood.

Neoplastic

1. *Diffuse lipoma of the cord*
 - It is rather rare.
 - The cord feels soft and lobulated.
 - The swelling is irreducible and has no impulse on cough.
2. *Malignant extension from the testis*
 - It may occur along its lymphatics.
 - The inguinal condition is, however, over-shadowed and explained by the scrotal one (i.e. the testicular tumor).